



Water Quality, Air Pollution, and Climate Change: Investigating the Environmental Impacts of Industrialization and Urbanization

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Abstract Human activities profoundly impact both biotic and abiotic components of ecosystems. Pollution, defined as the introduction of harmful substances into the environment, poses a significant threat to all living organisms. Even minimal concentrations of pollutants—whether in gaseous, liquid, or solid form—can disrupt ecosystem health. The advent of the Industrial Revolution marked the beginning of large-scale pollutant emissions, leading to severe repercussions for both human health and the environment. While the Industrial Revolution facilitated numerous advancements in services, science, and societal progress, it also introduced substantial environmental challenges. Today, the unprecedented levels of industrialization and urbanization have amplified global environmental concerns. Human-induced air pollution alone accounts for approximately 9 million deaths annually, presenting a critical public health threat. This research examines the environmental contamination arising from human activities, with a focus on water quality, air quality, and climate change. It aims to provide a comprehensive understanding of the environmental impacts of human actions, highlighting the pressing global issue of water pollution, particularly in developing countries. The study underscores the crucial role of effective

wastewater treatment in achieving sustainable development and explores how such measures can mitigate the impacts of climate change.

Keywords Air Quality · Atmosphere · Human Health · Pollution · Water Quality

1 Introduction

The "environment" refers to the complex and diverse settings in which a vast array of species, such as humans, animals, bacteria, and plants, live and interact. This environment is composed of several fundamental elements, each of which is indispensable for the sustenance of life and the preservation of ecological balance. These elements include land, which provides a foundation and resources for terrestrial life; water, which is essential for all forms of life and plays a key role in various biological and chemical processes; and the atmosphere, which offers the air we breathe and helps regulate the Earth's climate. Together, these components create a dynamic and interconnected system that supports the diverse forms of life on our planet and maintains the natural equilibrium necessary for their survival and well-being.

Pollutants, however, are defined as substances that are introduced into the environment in quantities exceeding their natural levels. These excess pollutants can disrupt the delicate balance of ecosystems and lead to detrimental effects on both the environment

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and the species that depend on it Briffa et al., 2020. The significant influence of human activities on the environment has led to extensive research focusing on the dynamic interactions between humans and their physical surroundings. These studies aim to understand how human actions affects the environment and how these impacts can be managed or mitigated to ensure sustainable development Peng & Jia, 2023

The environment is composed of both biotic and abiotic components. Biotic components include all living organisms, from the smallest microorganisms to the largest animals and plants. These living beings interact with each other and their surroundings in intricate ways. Abiotic components, on the other hand, consist of non-living elements such as the atmosphere, lithosphere (the earth's crust and upper mantle), and hydrosphere (all the water bodies on earth). These non-living elements provide the essential resources and conditions needed for life to thrive Huggett, 2023.

Understanding the interrelationship between biotic and abiotic components is crucial for environmental conservation and management. The interactions between living organisms and their non-living environment form the basis of ecosystems, which are vital for maintaining the earth's biodiversity and the overall health of the planet. Despite their harmful effects, human activities, contaminating water, air, and the environment at large Tumolo et al., 2020 produce toxic substances in solid, liquid, or gaseous forms. Over the past century, rapid industrial growth has improved living standards by offering a variety of services. However, this development has also led to the excessive exploitation of natural resources, increasing pollution levels. This phenomenon has been widely acknowledged as having a great deal of utility, but it also has significant drawbacks Crane-Kramer & Buckberry, 2023.

Pollution introduces a wide array of pollutants such as inorganic, organic, and organometallic chemicals, gases, nanoparticles, viruses, and radioactive isotopes into the environment, posing a recognized environmental hazard. These contaminants, discharged into the atmosphere, water bodies, and the overall environment, present a substantial risk to human health Aliyu & Amadu, 2017. In developing nations striving for economic advancement, water pollution emerges as a significant issue stemming from diverse sources such as industrial discharges, inadequate waste

management, agricultural runoff, oil spills, sewage, and wastewater Lin et al., 2022. This pollution leads to diminished oxygen levels, biodiversity loss, habitat destruction, and disruptions in aquatic ecosystems, profoundly affecting fish and other aquatic organisms. Consequently, aquatic populations suffer, and ecosystems face degradation risks.

Contaminated water also spreads waterborne diseases such as cholera, hepatitis, and dysentery, and has long-term effects on growth, development, reproduction, and cancer rates Alao et al., 2023. Additionally, economic losses from reduced tourism revenue and water cleanup costs result from pollution. To achieve sustainable development, strategies to prevent water pollution and effectively treat wastewater before disposal must be implemented Ayejoto et al., 2023. Given the constant movement of air, it can easily become contaminated and transfer pollutants from one place to another.

In our pursuit of sustainable development, it is imperative to implement targeted solutions that effectively reduce water pollution. Central to this effort is the deployment of efficient wastewater treatment systems, crucial for safeguarding our water resources from contamination Silva, 2023. Historical research from the 1980s uncovered a startling revelation: sulphur dioxide emissions, stemming from coal combustion in the Ohio Valley, travelled unprecedented distances via wind currents. This phenomenon led to the formation of acid rain, which had far-reaching consequences, affecting extensive regions of the eastern United States and Canada, stretching thousands of kilometres beyond its origin in the Ohio Valley Grenfelt et al., 2020. A comparable scenario unfolds in Asia, where powerful winds carry industrial pollutants from China across the vast expanse of the Gobi Desert. These pollutants accumulate during their journey, resulting in the formation of substantial yellow dust storms that adversely affect regions such as Japan and the Korean Peninsula. These storms not only diminish visibility but also pose significant risks to human health Agbasi et al., 2023.

Furthermore, stagnant air conditions created by high-pressure systems exacerbate air pollution. This stagnation allows pollutants such as emissions from factories and vehicle exhaust to concentrate at hazardous levels, increasing the likelihood of carbon monoxide poisoning Hayasaka et al., 2019. The "greenhouse effect" refers to the natural process by which certain

gases trap heat within the Earth's atmosphere, thereby maintaining a temperature conducive to life Ayejoto et al., 2022. However, since the Industrial Revolution in the mid-nineteenth century, human activities have significantly intensified this effect, leading to global warming. This anthropogenic climate change has disrupted natural cycles and weather patterns, resulting in severe consequences such as prolonged droughts, elevated temperatures, rising sea levels, increased incidences of flooding, and more frequent and intense storms Adamo et al., 2021, Ayejoto & Egbueri, 2023.

In addition to human activities, the ocean plays a crucial role in climate dynamics by absorbing and redistributing heat, carbon, and freshwater across the globe. Accurate simulations of oceanic processes are essential for understanding how these dynamics interact with and respond to human-induced changes in the atmosphere Ezugwu et al., 2019. Furthermore, the global movement of people has had profound effects on ecosystems and has contributed to shifts in climate patterns Semenza & Ebi, 2019.

The primary objective of this review is to explore the environmental impacts of human activities, particularly focusing on water and air quality as well as

climate change. Our study seeks to provide a comprehensive understanding of how human actions influence the environment and underscore the critical role of effective wastewater treatment in promoting sustainable development and mitigating climate change impacts. Given the widespread nature of water pollution, especially in developing nations, our findings emphasize the urgent need to invest in advanced wastewater treatment technologies and infrastructure. These investments are essential not only for preserving water quality but also for ensuring the resilience of ecosystems and communities in the face of ongoing environmental challenges.

2 Review Sources

The research papers are sourced from a variety of databases, ensuring a comprehensive collection of relevant literature. Figure 1 provides a detailed visual representation of these databases, illustrating the diversity and scope of the sources utilized in this study.

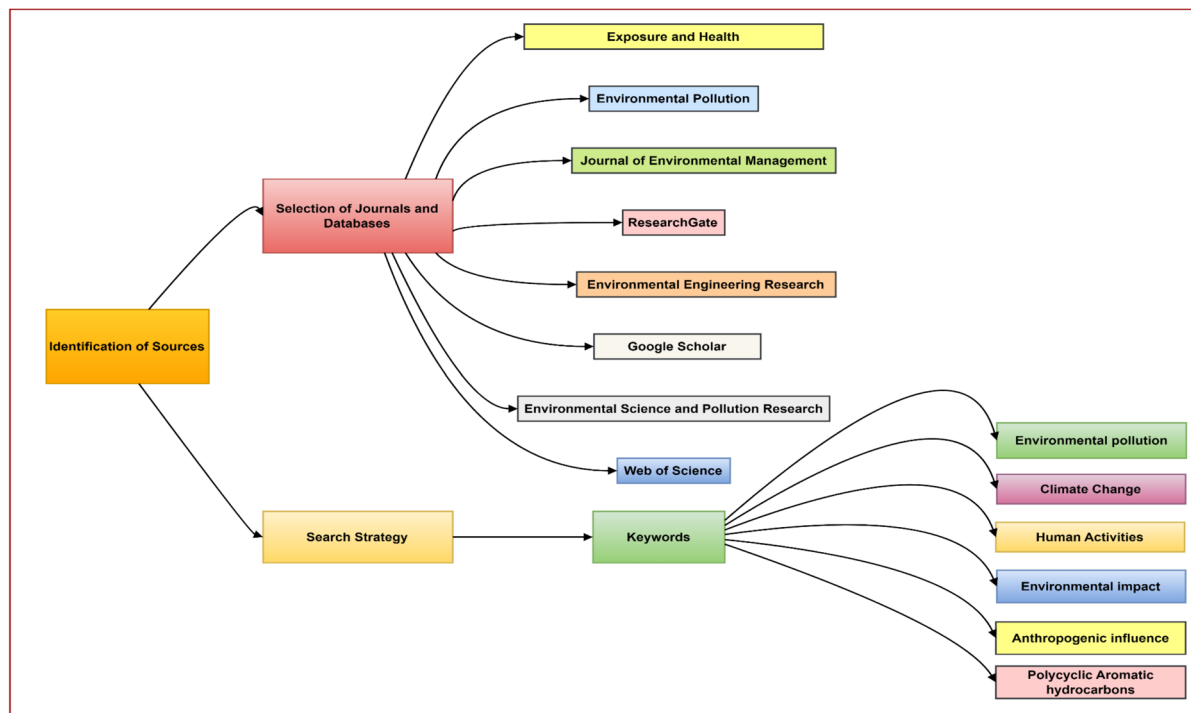


Fig. 1 Research methodology

3 Comprehensive assessment on water pollution

The introduction of toxic compounds into surface water (such as lakes, rivers, and oceans) or subsurface water (such as aquifers) is known as water pollution. These contaminants disrupt natural ecosystem processes or diminish the availability of clean water for beneficial uses Poikane et al., 2020. Energy emissions, including heat and radiation, can also contribute to water pollution Sathya et al., 2022, alongside the discharge of chemicals, waste, or pathogens into water bodies. Human health suffers significantly from exposure to unsafe water. Approximately 300,000 children under five years old die annually due to diarrhoea accounting for about 5.3% of total deaths Solomon et al., 2021. Additionally, studies in Palestine indicate a higher incidence of diarrhoea among those who consume untreated municipal water compared to those using desalinated or home-filtered water Mustafa et al., 2023.

Water is indispensable for a multitude of industrial, environmental, and physiological processes. It plays vital roles such as acting as a solvent, lubricant, temperature regulator, metabolite, and providing a habitat for countless organisms Arora et al., 2021. Despite its essential functions, water quality is often compromised when pollutants from various human activities exceed acceptable standards, making it unsuitable for its intended uses Ewuzie et al., 2021. This contamination is particularly concerning given water's crucial importance to all living organisms, including humans. Polluted water poses severe threats to environmental integrity, disrupting ecosystems and endangering human health. The specific impacts of these pollutants can vary widely depending on their nature and source Silva, 2023.

Moreover, the issue of water contamination is exacerbated by the looming global water scarcity crisis. Projections indicate that by 2025, five billion people out of an estimated global population of eight billion will be living in regions facing water scarcity Liu et al., 2024. This alarming forecast underscores the urgent need for sustainable water management practices to ensure that water remains available and safe for all its essential uses, thereby protecting both ecological balance and human health. The release of industrial and chemical substances poses one of the

most significant environmental challenges humanities faces today. Although many of these pollutants are present in trace amounts, their potential for causing significant toxicological effects cannot be overlooked. This risk is amplified when these pollutants interact in complex combinations, creating a cocktail of harmful substances.

As noted by Schwarzenbach et al., 2006, the presence of these pollutants in our water systems is widespread and growing. Table 1 provides a comprehensive overview of common water pollutants and their associated impacts. It includes pharmaceuticals, which can disrupt aquatic ecosystems even at low concentrations; industrial chemicals, which often persist in the environment and accumulate in the food chain; pesticides, which can be toxic to a wide range of organisms; steroid hormones, which can interfere with the reproductive systems of aquatic life; and other emerging substances that are increasingly being detected in water sources.

The global significance of freshwater contamination as a serious public concern should not be underestimated Narwal et al., 2023. For a comprehensive understanding of the various challenges related to sewage and waste management resulting from human activities, it is recommended to explore extensive resources on waste disposal and solid waste management. These resources provide detailed information on the methods and practices used to address and mitigate the impact of waste on the environment, including the collection, treatment, and disposal of sewage, as well as the management of solid waste materials Englande et al., 2015. Water pollution can arise from point sources or nonpoint sources. Point sources, like discharge pipes from municipal or industrial sewage systems, localize pollutants to specific areas, simplifying management in comparison to diffuse sources Babcock-Jackson et al., 2023. Nonpoint sources, like runoff from agricultural fields, pose a broader challenge as pollutants enter water systems over expansive and unconfined areas.

While advancements in sewage treatment technologies have improved management of diffuse pollution sources, significant challenges remain Fu et al., 2024. Figure 2 depicts the various origins of water pollution and underscores the complexity of controlling non-point sources.

Table 1 Impact of water pollution

Pollutant	Non-biodegradable organic matter	Inorganic dissolved solids	Biodegradable organic matter	Nutrients	Pathogens	Suspended solids
Reduced Oxygen Transfer	✓					
Toxicity	✓					
Foam Detergents	✓					
Noxious Scents	✓					
Pesticides	✓					
Plant Harm (Salinity)		✓				
Plant Toxicity (Ions)		✓				
Soil Permeability Issues		✓				
Non-biodegradability	✓					
Oxygen Depletion			✓			
Sewage Conditions			✓			
Fish Deaths			✓			
Excessive Algae				✓		
Ammonia Toxicity				✓		
Newborn Illnesses				✓		
Contaminated Groundwater				✓		
Water-borne Illnesses					✓	
Appearance Issues						✓
Sludge Plaques						✓
Adsorption of Contaminants						✓
Pathogen Protection						✓
Biological Oxygen Demand			✓			
Conductivity		✓				
Coliforms					✓	
Helminth Eggs					✓	
Phosphorus				✓		
Nitrogen				✓		
Total Dissolved Solids		✓				
Suspended Solids						✓
References	Mustafa et al., 2023	Silva, 2023	Narwal et al., 2023	Babcock-Jackson et al., 2023	Xie et al., 2022	Jerin et al., 2022

Despite ongoing efforts, there is optimism for future breakthroughs in addressing these challenges

3.1 Causes of Water Contamination

3.1.1 Domestic Sewage

Sewage collected from residential areas harbours a considerable concentration of pathogens, which are microorganisms responsible for causing various diseases. These pathogens are primarily expelled through human feces, making sewage from urban

environments particularly rich in potentially harmful germs. If these pathogens were to leak into natural water bodies, such as lakes or rivers, they could directly threaten public health Cui et al., 2019.

Another significant issue associated with sewage is the presence of organic chemicals that can decompose over time. These biodegradable substances are naturally found in sewage and are broken down by bacteria and other microorganisms present in

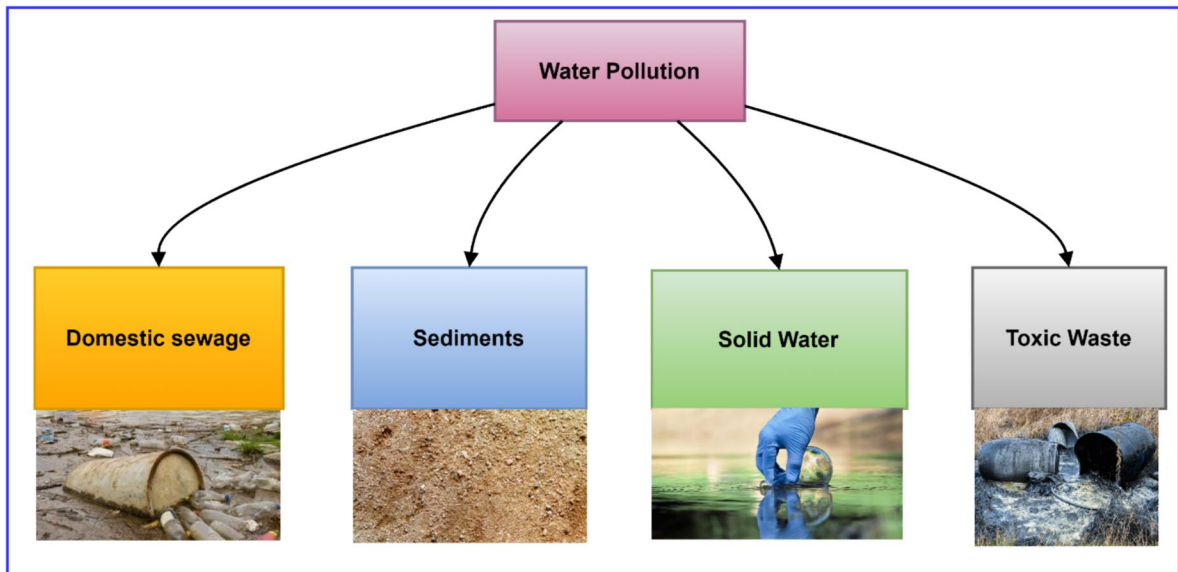


Fig.2 Sources of water pollution

water. However, this process of decomposition consumes dissolved oxygen from the water, leading to a decrease in oxygen levels Xie et al., 2022.

The discharge of untreated sewage into lakes and streams poses a significant risk to aquatic ecosystems. Fish and other aquatic organisms depend significantly on sufficient oxygen levels in the water for their survival and well-being. The introduction of sewage can disrupt this balance by reducing oxygen availability, thereby endangering the health and sustainability of aquatic life Nur-E-Alam et al., 2020.

In summary, sewage from households not only contains a variety of pathogens that can threaten human health if released into water bodies, but it also introduces organic compounds that degrade water quality by depleting oxygen levels essential for aquatic organisms. Proper management and treatment of sewage are crucial to mitigate these environmental and public health risks associated with urban wastewater disposal.

3.1.2 Solid Waste and Sediments

Improper disposal of solid waste poses a significant threat to water quality by contributing substantially to contamination. This issue encompasses a wide array of solid waste types originating from various sources such as private households, commercial enterprises,

institutions, and industrial facilities. Among these wastes are garbage, rubbish, electronic waste, general trash, sediment washed off from surfaces by rainwater (surface runoff), and debris from construction and demolition activities Almusawi et al., 2022.

In some regions, the deliberate discharge of solid waste into water bodies occurs due to several factors. These include the lack of adequate infrastructure for effective waste management, limited resources allocated to waste disposal, and insufficient or poorly enforced legislative measures. This problem is particularly acute in economically disadvantaged nations where the capacity to manage solid waste is often overwhelmed Jerin et al., 2022.

The consequences of such practices are far-reaching. They disrupt the delicate ecological balance of aquatic ecosystems by reducing the penetration of sunlight into the water, which is essential for photosynthesis and the sustenance of aquatic life Blanchet et al., 2022. Moreover, when solid waste settles and accumulates on the seabed or riverbed, it can smother and suffocate bottom-dwelling species, exacerbating ecological harm Brulé et al., 2022.

Addressing the issue of improper solid waste disposal requires concerted efforts in developing comprehensive waste management infrastructure, allocating sufficient resources for waste disposal and recycling programs, and enforcing stringent

environmental regulations. By doing so, we can mitigate the adverse effects on water quality and safeguard the health and integrity of aquatic environments worldwide.

3.1.3 Toxic waste

Toxic waste comprises a wide range of substances that present considerable dangers to both the environment and human health. These substances span various categories, each with its own distinct hazards: they can be poisonous, emitting toxins that harm living organisms; radioactive, emitting harmful radiation; explosive, posing immediate danger if ignited; teratogenic, capable of causing birth defects in developing fetuses; carcinogenic, potentially leading to cancer development; mutagenic, capable of damaging genetic material; and bio-accumulative, building up within organisms as they consume contaminated prey, ultimately concentrating in higher levels of the food chain. Each of these characteristics underscores the complex and multifaceted risks associated with toxic waste, necessitating rigorous management and containment practices to safeguard ecosystems and public health Jaishankar et al., 2014.

Examples of sources for these hazardous chemicals are widespread. They include pesticides like chlor-dane, dieldrin, and heptachlor, which are carried into water bodies through surface runoff from agricultural fields and residential lawns. Additionally, industrial plants and chemical processing facilities discharge wastewater containing toxic metals such as lead, mercury, and chromium, contributing further to environmental contamination Lalah et al., 2022.

Land pollution, another significant concern, introduces debris and rubble into water systems through various means like wind, animal transport, or precipitation. This process directly contaminates water bodies, and over time, pollutants can accumulate and eventually reach the oceans from inland waterways Shi et al., 2024. This pollution not only affects the aesthetic quality of aquatic environments but also poses immediate threats to specific species and entire ecosystems Zhang et al., 2024.

Furthermore, specific categories of solid waste, such as plastics and electronic waste, are acknowledged for their potential to degrade and emit harmful chemicals into the environment during decomposition. These materials are categorized as hazardous or

toxic waste due to their ability to release substances that can adversely affect ecosystems and human health over time. Proper management and disposal practices are essential to mitigate these environmental risks associated with the breakdown of such waste materials Abdel-Shafy & Mansour, 2018. This understanding underscores the critical need for proper disposal methods such as landfills, which are designed to contain and mitigate the risks associated with such waste materials.

3.1.4 Pathogens

Bacteria, viruses, protozoans, and helminths represent the primary types of organisms recognized for their potential to endanger human health through waterborne transmission. In the realm of water quality research, indicator organisms play a pivotal role due to the considerable difficulty and expense associated with detecting pathogenic organisms in water samples, which often contain low concentrations of these hazards. These bacterial markers are extensively utilized for their reliability in identifying the presence of fecal matter, which can harbor harmful microorganisms Motlagh & Yang, 2019.

Waterborne microorganisms have the potential to cause a range of illnesses in both humans and animals. Pathogens have been sporadically detected in surface waters contaminated by sewage or other sources. These pathogens have been associated with a variety of health problems in people, highlighting the critical importance of monitoring and maintaining water quality standards. Additionally, Schistosoma-type parasitic worms have been implicated in certain instances of waterborne diseases, emphasizing the diverse nature of health risks associated with contaminated water sources Richiardi et al., 2023.

3.1.5 Inorganic compounds

Inorganic water contamination encompasses several significant sources. One notable example is the presence of ammonia, which often originates from the waste products of food processing industries. These industries release ammonia as a byproduct of their operations, contributing to water pollution. Another concern is the introduction of heavy metals into urban storm drains by vehicles. These metals, such as lead and mercury, accumulate from automotive emissions

and runoff, posing risks to aquatic ecosystems upon entering water bodies. Furthermore, the application of acid mine drainage fertilizers can lead to the generation of phosphates and nitrates in water systems Yang et al., 2024a. These chemicals, essential for plant growth but harmful in excess, can contribute to nutrient pollution when they leach into waterways.

Additionally, sediment, known as silt, can be found in runoff from various human activities. These activities include slash-and-burn agriculture, construction projects, sewage systems, logging operations, and land clearance activities Adams et al., 2016. Sediment runoff poses challenges by clouding water and affecting aquatic habitats.

Freshwater salinization, caused by saline runoff, is another critical issue affecting freshwater ecosystems. This phenomenon refers to the pollution of freshwater bodies due to the introduction of salts from various sources. A significant contributor to salinity in freshwater systems is secondary salinization, also referred to as anthropogenic salinization, stemming largely from the extensive application of de-icing salts on roads. This phenomenon occurs when salts used to melt ice and snow on roadways eventually leach into surrounding soil and water bodies. The process is exacerbated by urbanization and infrastructure development, which increase the frequency and volume of salt applications. As a result, freshwater ecosystems experience elevated salinity levels, impacting aquatic life, water quality, and ecosystem health Cunillera-Montcusí et al., 2022. These salts enter water bodies through runoff and can alter the chemical composition of freshwater ecosystems, impacting aquatic life and water quality. For further insights into the complexities of nutrient pollution and its implications for water quality, please refer to the dedicated section on this topic.

3.2 Social Impact

The impact of human activities and our interactions with the natural environment have placed Earth's ecosystems in significant peril. Our actions, from industrialization to urbanization, agriculture, and resource extraction, have collectively threatened the delicate balance of ecological systems worldwide. These activities have resulted in habitat destruction, pollution, climate change, and the depletion of natural resources, all of which have profound implications for

biodiversity and the stability of ecosystems. Addressing these challenges requires concerted efforts in conservation, sustainable development, and global cooperation to ensure the resilience and health of our planet's diverse ecosystems for future generations. This peril stems largely from the disregard of traditional regulations and ethical considerations, resulting in widespread depletion and degradation of natural resources. This includes the deterioration of water quality and air purity, which are essential for sustaining life on our planet Gour et al., 2024. The pursuit of economic growth and an improved standard of living has driven these destructive practices, often without due consideration for their long-term consequences.

The evolution from a "sociology of environment" to "environmental sociology" reflects a broader academic and societal shift. It emphasizes the complex relationships between humans and their environment, aiming to guide human activities in ways that promote ecological health and sustainability Mihalache et al., 2024. This paradigmatic shift underscores the need for integrated approaches that consider environmental, social, and economic factors to ensure a balanced and resilient ecosystem.

Central to maintaining ecological balance is the diversity of organisms within ecosystems. From microscopic bacteria to large marine mammals, each species plays a crucial role in the intricate web of life. The interactions within these ecosystems are delicate and interdependent; any disturbance can have cascading effects throughout the entire ecosystem Weiland-Bräuer, 2021. For instance, the introduction of excessive chemicals into water bodies can fuel algal blooms, leading to oxygen depletion and the creation of "dead zones" where marine life cannot survive Iruoghene, 2022, Faquih et al., 2024. Such phenomena not only threaten aquatic biodiversity but also disrupt fisheries and other human activities dependent on healthy marine ecosystems.

Moreover, toxic algal blooms pose additional risks, releasing harmful neurotoxins that can affect marine mammals like whales and sea turtles, as well as human health in coastal communities. Industrial and municipal wastewater further exacerbates these issues, introducing pollutants such as heavy metals and chemicals into freshwater systems Dorantes-Aranda, 2023. These pollutants accumulate in organisms, reducing their lifespan and reproductive capacities, thereby endangering entire populations

and ecosystems Pasaoglu et al., 2021. Predators consuming contaminated prey, such as larger fish accumulating mercury from smaller fish, exemplify how human activities can lead to hazardous levels of pollutants in the environment Ritger et al., 2018. These pollutants not only affect marine life but also pose risks to human health through the consumption of contaminated seafood. Beyond chemical pollutants, marine ecosystems face threats from physical pollutants like marine debris. Plastics and other waste materials can entangle, suffocate, or be ingested by marine organisms, leading to starvation, injury, or death Saravanan et al., 2024. This underscores the urgent need for global efforts to reduce plastic waste and other marine debris through improved waste management practices and international cooperation. The challenges facing our ecosystems are multifaceted and interconnected. Addressing these challenges requires a holistic approach that integrates scientific knowledge, policy interventions, and public awareness. By promoting sustainable practices and environmental stewardship, we can mitigate the impacts of anthropogenic activities and safeguard the biodiversity and ecological integrity of our planet for future generations.

A considerable volume of solid waste, comprising items such as plastic bags and soda cans, often enters the marine environment through storm drains and sewer systems, eventually making its way into the oceans. The consequences for marine life have been severe, affecting more than 200 different species. Abandoned fishing equipment and various other forms of waste Mahajan-Tatpate et al., 2024 exacerbates this impact.

Besides the accumulation of physical waste, another critical concern is the rising acidity levels in our oceans. This increase in acidity primarily stems from the absorption of around a quarter of the annual carbon emissions generated by burning fossil fuels. The repercussions of this phenomenon are extensive: it can interfere with the neurological processes of marine life and impede the growth of shells in various species, particularly shellfish. This acidification poses a serious threat to marine ecosystems, affecting their biodiversity and ecological balance Falkenberg et al., 2020.

The phrase "water pollution" covers a diverse array of pollutants that affect aquatic environments. These contaminants include various chemicals,

pathogens such as bacteria and viruses, and physical changes such as temperature fluctuations. Each type of pollutant can have distinct effects on water quality and ecosystem health, highlighting the complex and multifaceted nature of water pollution. Understanding and addressing these different sources of contamination are essential for safeguarding water resources and maintaining the ecological balance of aquatic ecosystems Nguyen et al., 2024, Saxena et al., 2022b. While many chemicals exist naturally in water at controlled levels, elevated concentrations of these natural substances can disrupt aquatic ecosystems by blurring the distinction between natural components and pollutants. Moreover, substances that deplete oxygen levels in water, originating from both human activities and natural sources like decaying plant matter, can further stress aquatic environments Gavrilescu, 2021. Turbidity reduces light penetration, potentially hampering the growth of aquatic plants and affecting the respiratory systems of certain fish species Syeed et al., 2023. This phenomenon underscores the interconnectedness of environmental factors and highlights the cumulative impact of human actions on marine and freshwater ecosystems.

3.3 Water pollution and human health

The consequences of water pollution are profound and extensive, exerting a significant influence on global public health and mortality rates. A notable study published in *The Lancet* revealed that in the year 2015 alone, water pollution was directly responsible for causing the deaths of approximately 1.8 million people Southerland et al., 2022. This staggering figure underscores the severity of the issue and highlights the urgent need for effective mitigation strategies. Beyond fatalities, contaminated water remains a pervasive source of illness for a staggering one billion individuals annually across the globe. This widespread health burden disproportionately affects impoverished communities, where proximity to major industrial polluters increases susceptibility to water-borne diseases Bashpa et al., 2024.

The primary culprits behind water contamination are waterborne pathogens—microscopic organisms like bacteria and viruses that originate from human and animal waste. These pathogens infiltrate drinking water sources, leading to diseases such as cholera,

typhoid fever, and giardiasis Gu et al., 2023. The problem extends beyond developing nations; even in affluent societies, accidental or illegal discharges from sewage treatment plants, alongside agricultural and urban runoff, introduce harmful microorganisms into waterways Mishra et al., 2023.

Additionally, specific diseases like Legionnaires' disease exemplify the risks associated with contaminated water sources. This severe form of pneumonia can be transmitted through water systems.

In terms of chemical contaminants, water pollution encompasses a wide array of hazardous substances. These include heavy metals like copper, lead, chromium, and arsenic, as well as microplastics, pharmaceutical residues, and biological contaminants. Furthermore, pollutants like nitrogen dioxide from insecticides and radioactive compounds like chloroform, which are by-products of disinfection processes, pose significant health risks due to their detrimental effects on human health Ateia et al., 2020.

Addressing water pollution requires concerted efforts across various sectors, including policy, industry practices, and public awareness. Effective management strategies must prioritize safeguarding water quality to protect human health and mitigate the widespread impacts of pollution on communities worldwide.

3.3.1 Arsenic

Globally, approximately 140 million people face a concerning problem: their drinking water exceeds the WHO initial safety guideline of 10 µg per liter (µg/L) for arsenic levels. This issue highlights a widespread challenge where communities are confronted with water quality issues that jeopardize public health and safety. This standard was established to protect human health from the toxic effects of arsenic. A considerable amount of this arsenic contamination is naturally occurring, stemming from geological formations that leach arsenic into groundwater. However, the problem is exacerbated by human activities. Industrial operations, such as mining and smelting, as well as agricultural practices, including the use of arsenic-based pesticides and fertilizers, contribute significantly to the increased levels of arsenic in water supplies Jiang et al., 2012.

The presence of arsenic in drinking water is a serious public health concern. Long-term exposure to

arsenic can lead to a range of health problems, including skin lesions, cancer, cardiovascular diseases, and neurological effects. The fact that so many people are exposed to unsafe levels of arsenic underscores the urgent need for stricter regulatory measures, improved water treatment technologies, and comprehensive monitoring of water sources. Efforts must also focus on mitigating the sources of contamination and providing affected communities with safe drinking water alternatives. This global challenge calls for coordinated action from governments, health organizations, and industries to protect vulnerable populations from the harmful effects of arsenic in drinking water Shaji et al., 2021.

The detrimental effects of arsenic exposure on human health are both profound and multifaceted, affecting a wide range of bodily systems. Extensive research by the International Agency for Research on Cancer (IARC) has established a clear and definitive link between arsenic exposure and an increased risk of developing various cancers. These include malignancies in critical organs, which are essential for vital bodily functions.

Furthermore, comprehensive studies conducted by the National Institute of Environmental Health Sciences (NIEHS) in the United States have shed light on the vast array of health risks posed by arsenic exposure. Chronic exposure to arsenic is associated with a multitude of adverse health outcomes. Respiratory problems, such as persistent coughing and reduced lung capacity, are common among those exposed to arsenic. Additionally, there is a heightened vulnerability to developing diabetes, a chronic condition that affects the body's ability to regulate blood sugar levels.

Arsenic exposure also poses significant risks to children, including developmental delays and impaired cognitive function, which can have long-lasting effects on their overall health and well-being. Cardiovascular health is similarly compromised, with increased incidences of ischemic heart disease, a condition characterized by reduced blood flow to the heart. Moreover, chronic kidney disease, which impairs the kidneys' ability to filter waste from the blood, is another serious consequence of prolonged arsenic exposure Parvez et al., 2013.

This dual perspective—highlighting both the widespread exposure to elevated arsenic levels and its extensive health consequences—underscores the urgent need for global initiatives to address water

quality issues and mitigate the risks posed by arsenic contamination.

3.3.2 Copper

Copper can potentially contaminate drinking water when it leaches from aging copper pipes, some of which have been in use for many decades. The widespread adoption of copper plumbing in residential buildings started in the 1930s and continued throughout the twentieth century, leading to its almost ubiquitous presence Montagnino et al., 2022.

Excessive exposure to copper can result in various symptoms, including abdominal pain, vomiting, and gastrointestinal distress. These symptoms highlight the importance of monitoring copper levels in drinking water to ensure it remains within safe limits.

Fortunately, modern copper piping used today is designed to minimize the release of any harmful elements into the water supply. As a result, it is generally considered safe for use in residential plumbing systems Saxena et al., 2021a. This advancement underscores ongoing efforts to improve water quality and safety standards in plumbing infrastructure.

3.3.3 Lead

Over the course of many years, aging pipes have been found to potentially contain lead, which can serve as a direct pathway for this harmful element to enter the water supply Renner, 2010. Exposure to lead-contaminated water can lead to a range of neurological symptoms. These may include headaches, fatigue, muscle weakness, hallucinations, and in severe cases, seizures. These health effects underscore the serious impact of lead contamination on human health, particularly emphasizing the need for clean and safe drinking water to prevent such harmful outcomes Mason et al., 2014. Moreover, exposure to lead can lead to other health complications such as kidney disease, anemia, and severe gastrointestinal discomfort, often referred to as lead colic Wani et al., 2015.

The World Health Organization (WHO) has provided alarming estimates indicating that around 240 million individuals globally are exposed to water contaminated with dangerous concentrations of lead. Lead, a toxic metal, poses significant health risks, particularly when present in drinking water, leading to both short-term and long-term health complications.

WHO's comprehensive investigations highlight the severe impact of lead exposure, revealing that the detrimental effects result in over 850,000 deaths each year. These deaths are attributed to a range of health issues caused by lead toxicity, including neurological disorders, cardiovascular problems, and kidney damage, underscoring the urgent need for global action to mitigate lead contamination in water sources and protect public health Saxena et al., 2021b. This highlights the significant public health impact of lead contamination in water sources globally.

3.3.4 Chromium

The contamination of water by chromium is a critical concern, primarily arising from its natural leaching from rocks and soil, as well as its discharge from various industrial activities. When chromium is released into the environment through these processes, it can enter water systems, leading to significant pollution. Among the different forms of chromium, hexavalent chromium (Cr [VI]) is particularly worrisome. This form is largely produced through human activities such as manufacturing, mining, and chemical processing. Hexavalent chromium is highly toxic and poses severe health risks to humans, including the potential to cause cancer and other serious health issues. Therefore, the presence of hexavalent chromium in water sources represents a major environmental and public health challenge Tumolo et al., 2020.

Numerous industries play a role in the spread of hexavalent chromium [Cr (VI)], a hazardous and carcinogenic compound. Key contributors include the stainless-steel industry, where Cr (VI) is used in alloying processes to enhance corrosion resistance. The welding sector also releases Cr (VI) during metal fusion and cutting operations. Portland cement production, another significant source, involves the use of chromium compounds that can release Cr (VI) during the cement manufacturing process.

The leather tanning industry is a notable contributor, as it employs Cr (VI) in the tanning process to increase the durability and flexibility of leather products. Similarly, the textile dyeing industry uses chromium-based dyes, which can result in the release of Cr (VI) into the environment. In the chrome plating sector, Cr (VI) is essential for providing a hard, corrosion-resistant coating on various metal products Fan et al., 2024.

When people drink water contaminated with chromium, they can face a variety of serious health problems. One of the immediate effects is the appearance of skin rashes, as depicted in Fig. 3. Beyond skin issues, chromium exposure can lead to significant toxicity in vital organs such as the kidneys and liver, potentially causing long-term damage. It can also induce gastrointestinal discomfort, leading to symptoms like nausea, vomiting, and abdominal pain.

Moreover, chronic exposure to chromium increases the risk of developing cancer, as it is a known carcinogen. Men may experience reduced sperm quality, which can impact fertility. Additionally, prolonged ingestion of chromium-tainted water can contribute to the onset of anemia, a condition characterized by a deficiency of red blood cells or haemoglobin, leading to fatigue and other related symptoms. Thus, the health implications of chromium-contaminated water are extensive and severe, warranting immediate attention and remedial action to prevent such risks Saxena et al., 2021c. The absorption of chromium through water intake becomes significantly more pronounced when the concentration of total chromium in drinking water surpasses 25 µg per Liter (µg/L). This threshold

marks a critical point where chromium begins to pose a more substantial health risk due to its increased bioavailability. According to the WHO's the issue of chromium contamination affects a relatively small portion of the United States population, with less than one percent being exposed to potentially harmful levels. These guidelines underscore the importance of monitoring and regulating chromium levels in drinking water to protect public health and ensure the safety of water supplies Georgaki et al., 2023. This underscores the critical importance of monitoring and mitigating chromium levels in water sources to safeguard public health from these detrimental effects.

3.3.5 Perchlorate

Perchlorate contamination in water sources can arise from a variety of origins, encompassing both human industrial activities and natural environmental processes. In the realm of industry, perchlorate plays a crucial role as a chemical component in the manufacturing of numerous products and technologies. For example, it is integral to the production of slurry explosives, which are used in mining and construction

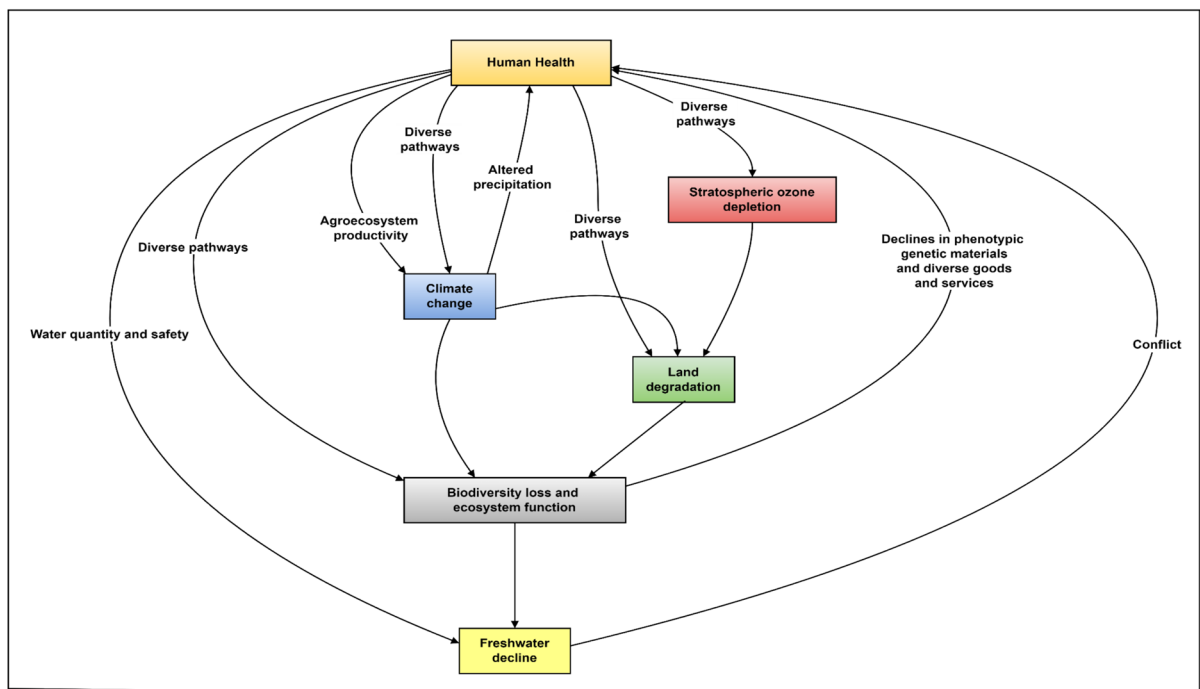


Fig.3 Relationship between varieties of environmental changes

for blasting purposes. Additionally, perchlorate is employed in airbag inflation systems, where it serves as an oxidizing agent that rapidly generates gas to deploy airbags in vehicles during collisions, thereby enhancing passenger safety Steinmaus, 2016.

Furthermore, perchlorate is a key ingredient in the formulation of road flares, providing the necessary oxidizer to sustain the intense, long-lasting flames required for signaling and emergency lighting. Its most notable industrial application, however, is in the development of solid rocket propellants. In this capacity, perchlorate contributes to the powerful chemical reactions that produce the thrust needed to propel rockets and missiles. These various uses underscore the significant role that industrial activities play in the introduction of perchlorate into water systems, adding to the contamination that can also originate from natural geological sources such as certain mineral deposits and atmospheric processes. Thus, addressing perchlorate contamination involves considering both its widespread industrial applications and its natural occurrences.

The reduction of thyroid hormones is particularly concerning during pregnancy and early childhood, as these hormones are vital for proper neurological development. Studies have demonstrated that insufficient thyroid hormone levels during these critical periods can adversely affect both intelligence and cognitive development Mughal et al., 2018. Thus, perchlorate contamination in water poses a serious risk to public health, particularly for vulnerable populations such as pregnant women and young children.

3.3.6 Fossil Fuel

Water pollution can originate from numerous contaminants produced during the extraction and refining of fossil fuels. This pollution includes a diverse array of harmful substances such as polycyclic aromatic hydrocarbons (PAHs), heavy metals like mercury and lead, benzene, methane, and other natural gases. Each of these pollutants poses a significant threat to water quality and public health. PAHs, which are organic compounds found in oil, coal, and tar deposits, can infiltrate water sources, leading to severe environmental and health impacts. Heavy metals, released during the mining and processing of fossil fuels, persist in aquatic environments, causing long-term contamination that can accumulate in the tissues of aquatic

organisms and, subsequently, in humans through the food chain. Benzene, a known carcinogen, and methane, a potent greenhouse gas, further contribute to water pollution, exacerbating the health risks associated with exposure. Howard et al., 2021.

A comprehensive study, published in 2003, highlights the scale of this issue. It reports that oil refineries worldwide release approximately 50,000 barrels, or 8 million liters, of petroleum by-products on a regular basis. This staggering amount of waste includes a variety of harmful substances that can severely degrade water quality. Alarming, around twenty-five percent of these by-products find their way into the water systems of certain countries. This infiltration of pollutants introduces a multitude of toxins into the water supply, exacerbating environmental and public health concerns Hussain et al., 2024.

The release of these contaminants into water bodies can lead to severe ecological and health repercussions. Polycyclic aromatic compounds and heavy metals, for instance, are known to persist in the environment and bioaccumulate, posing long-term risks to aquatic life and, ultimately, human populations that rely on these water sources. Benzene, a well-documented carcinogen, and methane, a potent GHG, further complicate the scenario by adding layers of both immediate and long-term health and environmental challenges.

In summary, the processing of fossil fuels not only contributes to air and soil pollution but also significantly threatens water quality through the release of a variety of hazardous substances. The study's findings underscore the urgent need for stricter regulations and innovative solutions to mitigate the impact of oil refinery waste on global water systems and public health.

3.3.7 Pharmaceutical Drugs

Pharmaceuticals encompass a wide array of active compounds that are used in various forms of medical treatments, including veterinary care, over-the-counter remedies, and prescription medications. These substances can be either artificially synthesized or naturally derived from sources like plants and animals.

The increasing use of pharmaceuticals in both agricultural practices and healthcare settings has raised concerns about their potential impact on the

environment Fernandes et al., 2021. These concerns stem from the fact that once pharmaceuticals are used, they can enter the environment through various pathways, such as wastewater from manufacturing facilities, hospitals, and residential use. As a result, pharmaceutical residues have been detected in surface water bodies and even in treated drinking water supplies Tudi et al., 2021.

A report published by the World Health Organization (WHO) in 2012 highlighted the presence of pharmaceuticals in drinking water, emphasizing the global nature of the issue and its potential implications for public health Schwartz et al., 2021. The concentrations of these pharmaceutical residues in water sources can range from very low levels, measured in nanograms per liter (ng/L), to slightly higher concentrations in areas affected by wastewater discharge, typically remaining below 100 ng/L Seo et al., 2023.

It's important to note that the concentrations of pharmaceuticals found in treated drinking water are usually well below levels that would pose any immediate health risks to humans Azanu et al., 2021. The current understanding among health authorities and environmental scientists is that exposure to such low concentrations of pharmaceuticals through drinking water is unlikely to cause adverse health effects. However, ongoing research and monitoring are essential to ensure that water treatment processes effectively remove these residues and that their potential long-term effects on ecosystems and human health are thoroughly understood.

In summary, while pharmaceuticals play a crucial role in modern medicine, their presence in the environment underscores the importance of responsible use and effective waste management practices to minimize potential impacts on water quality and public health.

3.3.8 Bio-Contaminants

A bio-contaminant is any living microorganism such as bacteria, viruses, or parasites that infiltrates water sources designated for drinking or daily use, thereby posing significant health hazards to humans Omarova et al., 2018. These harmful agents can include a variety of bacteria, such as *Escherichia coli*, which is infamous for causing gastrointestinal illnesses, and cyanobacteria, which can produce toxins leading to severe health issues. In addition to these bacterial

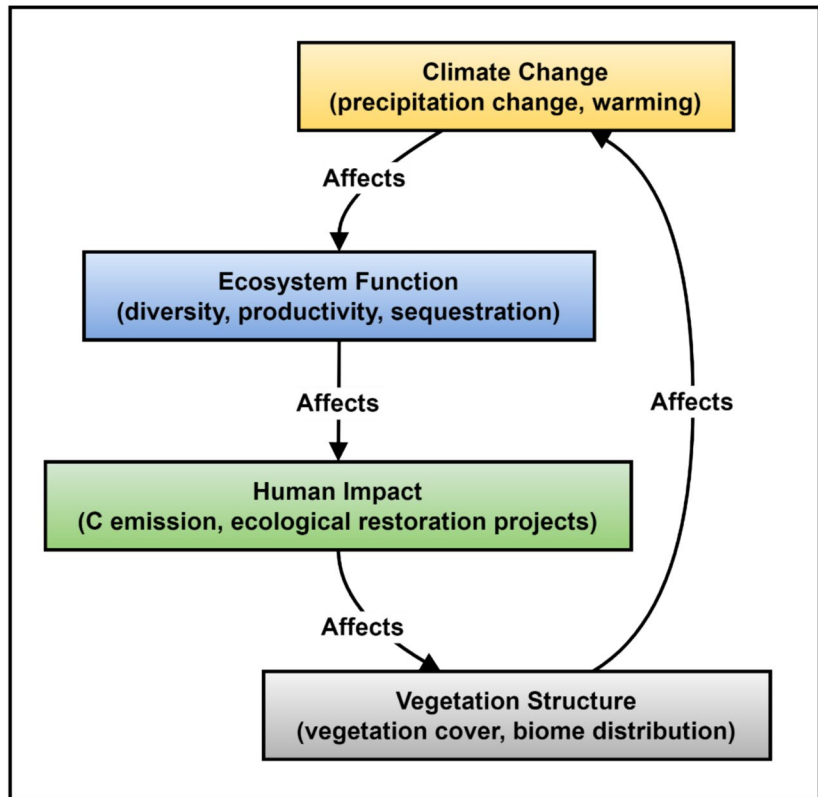
threats, parasites like *Giardia*, *Cryptosporidium*, and *Cyclospora* are known for contaminating water and causing serious waterborne diseases, leading to symptoms ranging from diarrhea to more severe health complications. Viruses, including adenoviruses and hepatitis A virus, also pose a significant threat to water safety, as they are capable of infecting water supplies and causing widespread health issues Bilung et al., 2017. The presence of these bio-contaminants in drinking water underscores the critical need for stringent water quality management and safety measures to protect public health.

The primary health impact of waterborne bio-contamination manifests most commonly as acute diarrheal disease, characterized by loose, watery stools often accompanied by vomiting and fever. This condition is widespread among children globally, affecting over 1.7 billion individuals annually according to estimates from the WHO. Tragically, this illness contributes to approximately 525,000 deaths each year, ranking it as the second leading cause of mortality among children worldwide.

In scientific research, the identification of the highly contagious *Cyclospora* species, particularly *Cyclospora cayetanensis*, has been a significant discovery. This pathogen was initially identified through rigorous laboratory investigations, with an illustration of this discovery provided in Fig. 4.

3.3.9 Pesticides

The use of pesticides in agricultural practices has the potential to contaminate water sources that are designated for human consumption. This contamination, primarily caused by herbicides, insecticides, and fungicides, presents a significant environmental and public health challenge Syafrudin et al., 2021. Studies have demonstrated that these chemical substances can lead to a wide array of detrimental health effects. For instance, exposure to these pesticides has been linked to an increased risk of developing various types of cancer. Additionally, there is evidence suggesting that these chemicals can impair neurodevelopment in infants, leading to long-term cognitive and developmental issues. Furthermore, pesticides have been associated with reproductive health problems, including reduced fertility and other reproductive disorders in men Wang & Qian, 2021. The widespread use of these chemicals in agriculture underscores the need

Fig. 4 Impact of climate change

for stringent regulations and effective measures to protect water quality and public health.

3.3.10 Nitrates

Nitrates are a class of chemical compounds that are created through a specific chemical process in which nitrogen atoms combine with oxygen atoms or ozone molecules. This reaction typically involves the oxidation of nitrogen, resulting in the formation of nitrate ions (NO_3^-).

In nature, this process often occurs through a series of steps in the nitrogen cycle. Initially, nitrogen in the atmosphere, primarily in the form of nitrogen gas (N_2), is converted into more reactive forms such as nitrogen oxides (NO and NO_2) through processes like lightning or microbial activity in the soil. These nitrogen oxides then react with oxygen or ozone (O_3) in the atmosphere, especially under the influence of sunlight and other environmental conditions, leading to the formation of nitrates Ng et al., 2017. These compounds naturally exist in the environment and

are frequently present in a variety of foods and food preservatives. Although they are found in nature, the predominant source of nitrates is the commercial fertilizers that are widely applied in modern agricultural practices. These fertilizers are extensively used to enhance crop yields and are a major contributor to the levels of nitrates we encounter in the food supply and the environment Singh & Saxena, 2021.

The European Union has established regulations to control the levels of nitrates and nitrites in drinking water, setting maximum allowable concentrations for nitrites Zendehbad et al., 2022. These limits are crucial to ensure public health and safety. When the concentrations of these compounds exceed the specified limits, it is essential to take immediate corrective actions to reduce them to acceptable levels. Such actions may include improving water treatment processes or reducing the use of nitrate-rich fertilizers in agriculture. Failure to adhere to these regulations can result in significant consequences for producers, including financial penalties Ward et al., 2018.

3.4 Climate change

To comprehensively understand climate change, it is essential to start by defining the term "climate." Climate can be described in two primary ways. This statistical description includes parameters such as temperature, precipitation, wind, and other related factors. Second, "climate" is often understood as the average weather conditions observed in a specific region over a long period. In essence, climate can be generally defined as the state of the climatic system, incorporating both the variability and averages of various atmospheric elements Koutsoyiannis, 2021.

Climate change, often referred to as global warming, encompasses the sustained elevation in Earth's average surface temperature over an extended period. This phenomenon primarily stems from human activities, notably the combustion of fossil fuels such as coal, oil, and natural gas. When these fuels are burned, they emit GHGs like carbon dioxide (CO₂) into the atmosphere. GHGs possess the capability to trap heat, contributing to the warming of the Earth's surface. They include significant environmental impacts such as heightened sea levels due to melting polar ice caps and glaciers. Moreover, climate change exacerbates the frequency and intensity of extreme weather events such as hurricanes, droughts, and floods. These consequences not only pose threats to ecosystems and biodiversity but also have profound socio-economic implications, affecting agriculture, water resources, and human health worldwide. Addressing climate change involves global efforts to reduce GHG emissions, adapt to its impacts, and foster sustainable practices to mitigate future risks to the planet and its inhabitants Saxena et al., 2023, Manikandan & Saraswathi, 2023, Wei et al., 2020.

A strong consensus exists among scientists worldwide that climate change is a real and urgent issue. Major scientific organizations, such as the National Aeronautics and Space Administration (NASA), the National Oceanic and Atmospheric Administration (NOAA), and the Environmental Protection Agency (EPA) of the United States, have confirmed that human activities are the primary drivers of recent climate change. These organizations highlight the profound impacts of climate change on natural systems, human health, and global economies.

The Geneva Report published by the World Health Organization (WHO) provides a detailed analysis

of how global climate change measures affect public health. Figure 3 from this report illustrates the comprehensive impact of climate change on human health, showing that the general population is significantly affected by these changes. Health impacts include increased incidence of heat-related illnesses, respiratory disorders due to poor air quality, and the spread of vector-borne diseases.

Despite various mitigation efforts, the Earth's climate system continues to experience significant stress and pressures. The ongoing emission of GHG and other pollutants exacerbates the challenges posed by climate change. As a result, it is crucial to implement effective strategies for reducing GHG emissions, enhancing resilience to climate impacts, and promoting sustainable development practices to protect both the environment and human health Skendžić et al., 2021.

The increasing frequency of floods and droughts, coupled with disruptions to our natural water ecosystems, present profound challenges to the sustainable coexistence of humanity with the Earth Bhaga et al., 2020. The far-reaching impacts of extreme climate events encompass a broad spectrum of effects with profound implications. These include elevated levels of air pollution, a rise in the prevalence of infectious diseases, and substantial transformations in both freshwater and marine ecosystems. These ecosystems play pivotal roles in ensuring food and water security, underscoring the interconnectedness between climate dynamics and the sustainability of vital resources Sandifer et al., 2015. Conversely, the observable impacts of climate extremes manifest through tangible alterations in our environment's physical characteristics, climate patterns, and weather systems. These transformations primarily result from human actions, influencing ecological equilibrium in ways that can either prove advantageous or pose challenges Saxena et al., 2022a.

While the Earth possesses a degree of resilience to environmental stresses, there are limits to this resilience. Human activities, particularly those that degrade natural systems and contribute to climate change, often exceed the Earth's capacity to regenerate and restore balance. Water contamination, a direct consequence of human-induced climate change, stands out as a pressing issue that demands immediate corrective measures to safeguard human health and ecological integrity Fallah Shayan et al., 2022.

Human actions profoundly influence all facets of the environment, including climate and weather dynamics. While some activities contribute positively to environmental health, others lead to degradation and disruption. The capacity of ecosystems to withstand damage is finite; thus, sustained human impacts threaten long-term ecological stability Arora et al., 2023, D'Amato et al., 2020.

The visible alterations in Earth's surface due to ongoing pressures on the climate system serve as stark reminders of the urgent need for comprehensive environmental stewardship and mitigation strategies Shi et al., 2023. In addition to the immediate concerns posed by rising surface temperatures, the sustainable future of human habitation on Earth faces significant challenges from the increasing frequency of extreme weather events such as floods and droughts. Furthermore, the integrity of our natural water systems, vital for biodiversity and human livelihoods, is increasingly compromised Hwabejire et al., 2023, Wang et al., 2023. These issues underscore the complex interplay between human activities, environmental health, and the future of life on our planet.

The World Health Organization's Geneva Report underscores a critical interplay between human activities and climate conditions, both of which have profound and simultaneous impacts on human well-being D'Amato et al., 2015. The actions of humanity exert substantial control over weather patterns, climate processes, and the overall health of the environment. Human activities, ranging from industrial emissions and deforestation to urbanization and agricultural practices, shape atmospheric conditions and alter natural climate cycles. These activities can lead to changes in precipitation patterns, temperature fluctuations, and the frequency of extreme weather events. Furthermore, they impact ecological balance, biodiversity, and the quality of air, water, and soil. Understanding and mitigating these impacts are critical for preserving ecosystems, safeguarding public health, and promoting sustainable development globally. Some of these activities contribute positively to ecological balance, while others pose potential risks and adverse outcomes Ugbune et al., 2023. Despite the environment's resilience to absorb certain levels of stress, the cumulative impact of harmful human practices often exceeds its capacity for recovery Pałczyński et al., 2018. For example, the pollution of water sources arises as an inevitable outcome

of climate change intensified by human activities, demanding immediate and specific actions to address the issue effectively. This situation underscores the critical need for prompt intervention strategies aimed at mitigating the environmental impact and safeguarding water quality for present and future generations Wikel, 2022.

In essence, while certain human endeavors contribute positively to environmental well-being, others underscore the urgent need for mitigation strategies. The environment's ability to withstand stress is finite, and actions to address the unavoidable repercussions of human-induced climate change, such as water contamination, are imperative for safeguarding human health and ecosystem integrity Sorensen et al., 2018. Thus, proactive measures are essential to navigate the delicate balance between human activities and environmental sustainability in the face of ongoing climate challenges.

3.4.1 Climate Change's Impact on Human Health

Air pollution stands as a formidable challenge in our modern era, primarily due to its dual impacts on climate change and human health, thereby affecting the overall well-being of societies White et al., 2023. Among the array of pollutants, particulate matter (PM) is a significant culprit, comprising fine particles that, when inhaled, penetrate deep into the respiratory system. These particles not only impair lung function but also pose risks to cardiovascular health and can detrimentally affect neurological and reproductive systems.

Ozone, while beneficial in the upper atmosphere for shielding against harmful ultraviolet radiation, becomes problematic at ground level. Here, it contributes to respiratory and cardiovascular issues when present in sufficient concentrations Bouchard et al., 2019. Additionally, VOCs and PAHs, for instance, are volatile compounds that pervade the atmosphere and can lead to various health complications upon exposure Saxena et al., n.d.

The immediate effects of carbon monoxide are also notable, capable of causing harm shortly after exposure if concentrations are high enough Archer et al., 2023. Furthermore, heavy metals such as lead present serious health risks, potentially causing acute poisoning or chronic toxicity depending on the level and duration of exposure Elsayed et al., 2023.

Human activities have a profound impact on the natural world, particularly through the contamination of essential resources such as soil, water, and air. The industrial revolution, while ushering in remarkable advancements in technology, society, and service provision, also brought about significant levels of air pollution, which pose direct threats to human health Eckelman & Sherman, 2016. This period marked a turning point where human influence on the environment began to manifest in ways that affect not just ecosystems but also the broader fabric of society, including legal frameworks, economic systems, and societal norms Saxena et al., 2024a.

The global challenge of environmental pollution has far-reaching implications, impacting ecosystems in ways that threaten food security, accelerate ice melting and species extinction, and jeopardize the survival of plant species critical for biodiversity and human livelihoods. Over time, these environmental changes can also lead to adverse effects on human health, particularly cardiovascular health Florentina & Io, 2011.

In addition to chronic environmental changes bring with them not only immediate physical risks such as injuries and disease outbreaks but also significant psychological and social challenges, including displacement and community disruption. For example, cities like New Delhi illustrate the detrimental effects of poor air quality on public health, highlighting the urgent need for effective environmental policies Echemdu et al., 2022.

Air pollution, a prominent factor in the degradation of our environment, presents pressing threats to human health by causing harm to both the respiratory and cardiovascular systems. The pollutants released into the air, such as particulate matter and toxic gases, have direct and detrimental effects on human health, leading to increased risks of respiratory infections, exacerbation of asthma and other respiratory conditions, and heightened susceptibility to cardiovascular diseases. Addressing air pollution is imperative not only for safeguarding public health but also for preserving the overall well-being of ecosystems and biodiversity Zazouli et al., 2023, Saxena et al., 2024b. Exposure to pollutants like ozone and particulate matter can lead to a range of health issues including chest pain, respiratory discomfort, and reduced lung function, underscoring the critical need for mitigation strategies and public health interventions [Fig. 4].

Climate change has far-reaching consequences for public health, exacerbating concerns related to allergies and severe allergic responses like asthma. This is compounded by rising temperatures, heightened levels of pollen, and extended periods conducive to plant growth. These factors collectively intensify the challenges faced by healthcare systems and individuals managing respiratory conditions, highlighting the urgent need for adaptive measures and enhanced healthcare strategies to mitigate these impacts Morantes et al., 2021. These environmental shifts not only heighten susceptibility to respiratory ailments. Particularly vulnerable are populations residing in regions unaccustomed to extreme heat, such as northern latitudes Schraufnagel, 2020.

Among those most at risk are outdoor workers, student athletes, and homeless individuals, who endure prolonged exposure to elevated temperatures Heo et al., 2025. Additionally, low-income households and elderly populations, often lacking access to air conditioning, face heightened health risks during heatwaves Bhattarai et al., 2024.

Moreover, climate change alters air quality both indoors and outdoors, worsening respiratory and cardiovascular conditions like asthma and difficulty breathing due to increased temperatures and erratic weather patterns Samuel et al., 2023. Future projections indicate a heightened frequency and severity of wildfires, which not only emit smoke but also release harmful pollutants into the atmosphere, further compromising respiratory health Yang et al., 2024b, Kang et al., 2024.

Due to healthcare access and economic inequities, disease-carrying ticks and mosquitoes are spreading worldwide, disproportionately affecting vulnerable communities in less developed regions Kumar et al., 2022. These disparities underscore the differential risks faced by diverse racial and ethnic groups worldwide. Climate change disproportionately affects the health of women and girls more severely than men and boys in many regions, despite variations in impact across different contexts. This disparity stems from several factors: women and girls are often found in lower socioeconomic positions and are more likely to adhere to traditional gender roles, which can limit their ability.

To understand the broader implications on mental health, researchers have examined various indicators such as mortality rates, suicide rates, incidents

of self-harm, and admissions to psychiatric hospitals Owheruo et al., 2023, Jeong et al., 2024. Vulnerable populations include individuals with a history of mental illness, indigenous peoples, immigrants, refugees, children, adolescents, and the elderly. These groups are particularly susceptible to the psychological toll of climate change, experiencing emotions ranging from profound ecological grief and despair to pervasive worry. Such emotional responses are considered natural reactions to the deteriorating state of the natural world and may prompt behaviours aimed at adaptation and mitigation.

The geographical spread of disease-carrying ticks illustrates another consequence of climate change. Rising temperatures expand the habitats suitable for ticks, which are vectors for diseases such as Lyme disease and tick-borne encephalitis Chu & Karr, 2017. This expansion is expected to increase disease prevalence in countries across the Northern Hemisphere. For example, projections indicate that a 2-degree Celsius increase in temperature could lead to earlier and prolonged Lyme disease seasons, along with a substantial rise in cases in the United States over coming decades.

Similarly, climate change is anticipated to exacerbate outbreaks of diseases like cholera, caused by vibrio bacteria, due to changes in sea surface temperatures and salinity Costa et al., 2023. Certain demographic groups face heightened health risks due to climate change. Moreover, people living in remote rural areas, often lack access to resources and infrastructure that could help them cope with or mitigate the impacts of extreme weather events such as hurricanes, droughts, bushfires, and rising sea levels Xiaoju et al., 2024, Adla et al., 2022.

In conclusion, climate change not only poses significant environmental challenges but also has profound implications for public health, with differential impacts that disproportionately affect vulnerable populations. Addressing these disparities requires comprehensive strategies that include adaptation measures, equitable access to healthcare and resources, and policies aimed at reducing GHG emissions.

4 Comprehensive Assessment on Water Pollution

Air pollution from natural disasters like forest fires and volcanic eruptions was already there long before

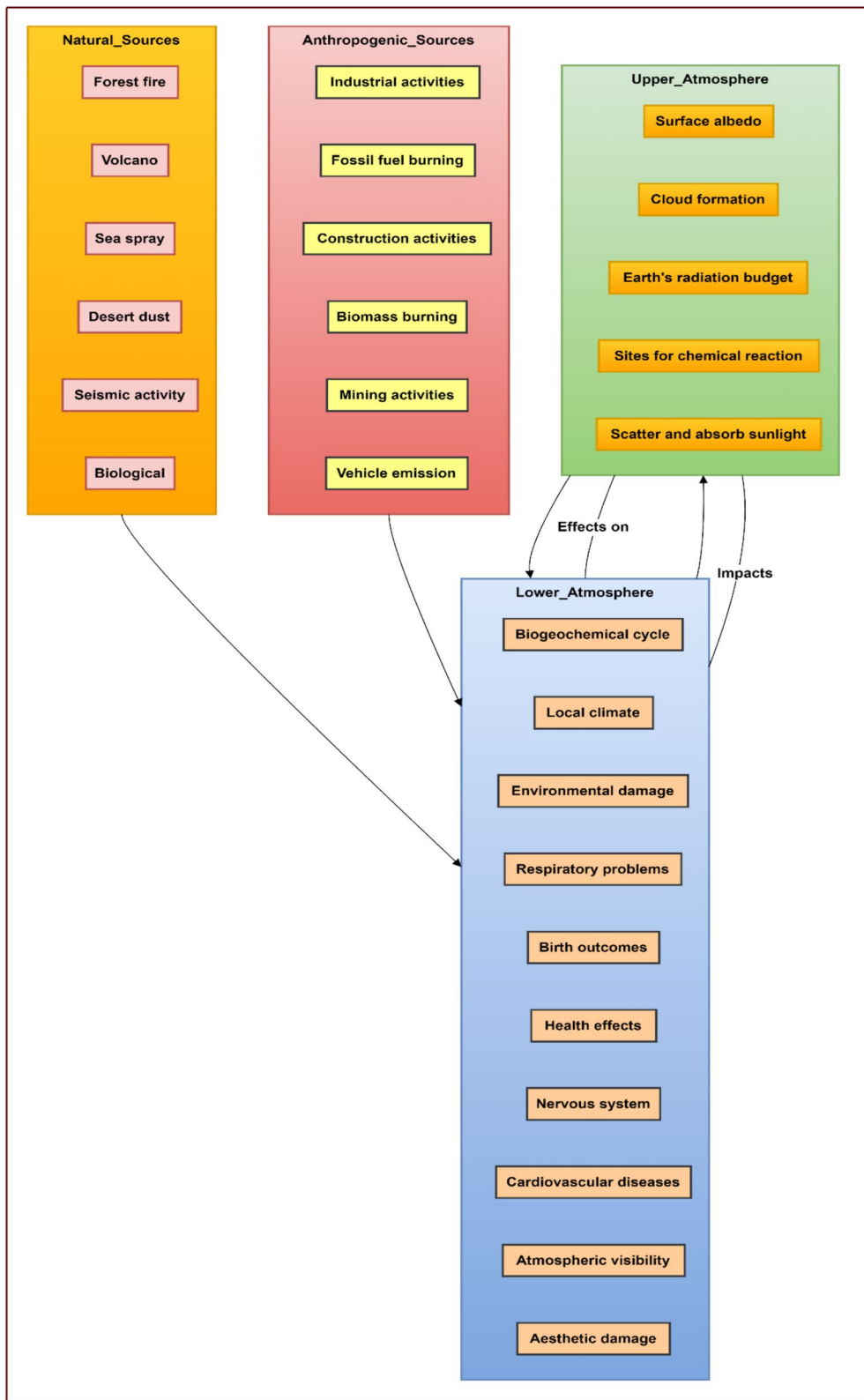
humans appeared on the scene. However, the contemporary environmental crisis we face today predominantly stems from human activities that have intensified air pollution Wikel, 2022. This escalation began with the widespread use of coal, which introduced GHG into the atmosphere, precipitating profound consequences for ecosystems and all forms of life. The impact of air pollution extends beyond humans, affecting animals, crops, infrastructure (via acid rain), and natural habitats (through climate change, ozone depletion, and habitat degradation) Ghorani-Azam et al., 2016.

Air pollution, arising from both natural phenomena and anthropogenic activities, poses significant health risks, primarily targeting the respiratory and cardiovascular systems of individuals. The severity of these health impacts varies widely among people due to factors such as the type and concentration of pollutants, individual health conditions, and genetic predispositions Kim et al., 2020.

In addition to the emissions caused by human activities, natural sources also contribute significantly to air pollution through various mechanisms. Methane, for instance, is released naturally during the decomposition of organic matter in soils. Volcanic activities emit ash and gases into the atmosphere, which can contain sulfur dioxide and other pollutants. Wildfires, which can occur naturally or be exacerbated by human actions such as land clearing, release substantial amounts of smoke and particulate matter.

These natural processes interact dynamically with human-induced emissions in the atmosphere. They create complex feedback loops where pollutants can interact and undergo chemical reactions, leading to secondary pollutants like ozone and fine particulate matter. These interactions not only contribute to localized air quality issues but also have global implications for climate change. For example, certain pollutants, such as methane and black carbon from wildfires, can contribute to the greenhouse effect and influence regional and global climate patterns Edo et al., 2023.

It is imperative to address the shared sources of these challenges, particularly the emissions resulting from the combustion of fossil fuels (as illustrated in Fig. 5), in order to effectively mitigate the impacts of air pollution and climate change. Implementing sustainable environmental practices is crucial for safeguarding human health and the well-being of all



◀**Fig. 5** Sources of air pollution and health impact

living organisms amidst these interconnected environmental challenges.

4.1 Structure of Atmospheric Pollution

Climate change and air pollution are significant issues that pose serious threats to the global environment and human health. The complex interplay between these two challenges highlights their collective impact on our planet. Urban areas are especially burdened by high levels of major air pollutants, including sulfur dioxide, nitrogen dioxide, and carbon monoxide. These harmful substances primarily originate from the burning of fossil fuels such as fuel oil, gasoline, and natural gas. This combustion process takes place in power plants, vehicles, and a variety of industrial activities. The high concentration of these pollutants in cities contributes to deteriorating air quality, which in turn exacerbates health problems and environmental degradation Roberts, 2021.

Ozone, a reactive gas and a major component of smog, forms in the atmosphere through intricate chemical reactions involving nitrogen dioxide and volatile organic compounds, including those emitted from gasoline vapours containing carbon monoxide—often termed the "silent killer" Edo et al., 2022. Carbon monoxide, despite being both odourless and colourless, poses a substantial risk to human life by interfering with the bloodstream's oxygen-carrying capacity. In high concentrations, carbon monoxide can completely obstruct oxygen transport, potentially leading to fatalities.

The combustion processes responsible for emitting these pollutants also release other hazardous substances into the surrounding environment. When combined with precipitation, these pollutants contribute to the formation of acid rain, which poses significant threats to sensitive ecosystems such as lakes and forests. The detrimental effects of acid rain extend to endangering aquatic life, harming vegetation, and altering soil chemistry, thereby impacting entire landscapes.

In summary, the complex interactions between emissions from fossil fuel combustion, the formation of ozone and other pollutants, and their culmination

in acid rain illustrate the interconnected challenges of climate change and air pollution. Addressing these issues requires integrated efforts in policy, technology, and global cooperation to mitigate their far-reaching impacts on both the environment and human health.

4.1.1 Ozone (O₃) Impact in the Atmosphere

Ozone (O₃), a molecule composed of three oxygen atoms, exists in varying concentrations near the Earth's surface as well as in the upper atmosphere. When found at ground level, particularly on hot and sunny days, ozone becomes a significant concern for human health. It forms a crucial part of smog, a complex mixture of pollutants, and is officially classified as a hazardous air pollutant due to its potential adverse impacts on human health and the environment.

Exposure to elevated levels of ground-level ozone can lead to a range of health problems. These include respiratory issues such as aggravated asthma, decreased lung function, and increased susceptibility to respiratory infections. Additionally, ozone exposure can exacerbate existing cardiovascular conditions and cause inflammation of the airways, leading to discomfort and reduced quality of life. The severity of these health effects depends on factors such as the duration and intensity of ozone exposure, as well as individual susceptibility Bradu et al., 2022.

Particles smaller than 10 µm can reach the lungs and potentially enter the bloodstream, causing irritation and inflammation of the airways. This can exacerbate respiratory conditions, and may increase the risk of respiratory infections Edo et al., 2024a.

Therefore, both ground-level ozone and particulate matter pose significant health risks, especially to vulnerable individuals with pre-existing respiratory conditions. Understanding these pollutants and their effects is crucial for implementing effective air quality management strategies to protect public health and the environment Upadhyay, 2020.

Table 2 highlights ozone's contrasting roles: stratospheric ozone acts as a vital UV shield, while ground-level ozone, formed by photochemical reactions, poses health risks like respiratory problems and environmental harm such as reduced agricultural productivity. It underscores the need for strategies to mitigate ground-level ozone pollution.

Table 2 Comparative impact of Ozone on the atmosphere and human health

Parameter	High Atmosphere (Stratosphere)	Ground-Level (Troposphere)	Reference
Formation	Naturally occurs, forming the ozone layer	Forms from VOCs and NO _x reactions under sunlight	Zhang et al., 2023, Gautam et al., 2023
Primary Role	Protects against harmful UV radiation	Component of smog, contributing to pollution	Zhang et al., 2023
Health Impacts	Beneficial (no direct health impacts)	Respiratory issues: lung function decline, asthma exacerbation, COPD risk	Gautam et al., 2023
Short-Term Health Effects	None	Chest pain, coughing, throat irritation	Gautam et al., 2023
Long-Term Health Effects	None	Chronic respiratory diseases, lung function deterioration	Zhang et al., 2023, Gautam et al., 2023
Environmental Impacts	Vital for blocking UV light	Damages vegetation, reduces crop yields	Zhang et al., 2023
Economic Costs	Positive (prevents skin cancer & cataracts)	Healthcare costs, agricultural losses	Zhang et al., 2023
Climate Change Role	Minor warming effect (as a greenhouse gas)	Contributes to smog, indirectly reducing CO ₂ absorption by plants	Zhang et al., 2023, Gautam et al., 2023
Regulatory Standards	Not required	Maintained below 70 ppb (EPA standard)	Zhang et al., 2023

Individuals who are generally healthy may experience some of these effects discussed; however, those who suffer from respiratory issues are particularly susceptible to more severe consequences Ebi et al., 2021. This susceptibility can lead to heightened use of pharmaceuticals, increased visits to healthcare providers and emergency rooms, more frequent hospital admissions, and greater absences from school or work.

Exposure to oxygen has been identified as a trigger for asthma flare-ups and is implicated in the onset of asthma itself. Moreover, studies have shown a correlation between elevated oxygen levels and increased mortality rates from respiratory disorders in regions with high concentrations of this gas Edo et al., 2024b.

Among the various air pollutants, fine particulate matter (PM_{2.5}) and ground-level ozone (O₃) are of primary concern Bălă et al., 2021. These pollutants have been found to harm agricultural crops (O₃), impair visibility (PM_{2.5}), and heighten the vulnerability of certain tree species to diseases (O₃) Montes et al., 2022. Smog, the atmospheric phenomenon predominantly composed of ozone (O₃), forms when other airborne contaminants react under sunlight Wróblewska & Jeong, 2021.

4.1.2 Carbon monoxide

Carbon monoxide (CO) is a dangerous gas that is both colorless and odorless, making it undetectable by human senses. It is produced when carbon-containing fuels like natural gas, gasoline, and wood undergo incomplete combustion. This can happen in various settings, including vehicles, power plants, wildfires, and incinerators, leading to the release of CO into the atmosphere. Vehicles, particularly in urban areas, are significant contributors to outdoor CO emissions. Traffic congestion exacerbates this issue, causing high levels of CO pollution. Power plants that burn fossil fuels and wildfires also add to CO emissions. Incinerators used for waste disposal can emit CO when the combustion process is incomplete. The presence of CO in the atmosphere is a major environmental and public health concern. Efforts to reduce CO emissions focus on improving combustion efficiency, regulating industrial processes, and reducing traffic congestion to mitigate the harmful effects of this invisible gas Sheoran et al., 2022.

In addition to direct sources of combustion, other atmospheric compounds, including methane, non-methane hydrocarbons, and various organic molecules found in surface waters and soils, can also

contribute to the production of carbon monoxide (CO) through photochemical reactions. These reactions occur when sunlight interacts with these compounds, leading to a series of complex chemical processes that ultimately result in the formation of CO. Methane and non-methane hydrocarbons, when exposed to ultraviolet light, can break down and participate in reactions with other atmospheric components. Similarly, organic molecules in surface waters and soils can undergo transformations under the influence of sunlight, adding to the CO levels in the atmosphere. This highlights the multifaceted nature of CO production, extending beyond just combustion to include photochemical processes involving various organic substances Liu et al., 2022. Natural events such as wildfires and industrial processes that involve burning fuels other than those used in transportation further add to the atmospheric burden of CO. The cumulative impact of these industrial activities on overall CO levels is noteworthy.

Indoors, internal combustion engines and emission control systems in vehicles operate less efficiently when cold, resulting in higher levels of CO emissions. This inefficiency occurs because cold engines have lower air-to-fuel ratios, which leads to incomplete combustion. As a result, catalysts within emission control systems also take longer to reach optimal operating temperatures, further contributing to increased CO output Dey & Dhal, 2019.

While instances of extremely high CO concentrations are rare, they can occur under specific conditions influenced by emissions patterns, weather conditions, and geographical factors. CO is known for its stability in the atmosphere, with a typical lifespan ranging from two to four months Wallington et al., 2022, Maciejczyk et al., 2021. Its stability, coupled with its lack of distinctive smell or taste, complicates its detection and diagnosis. Symptoms of CO poisoning can mimic those of common illnesses, posing challenges in identifying and treating cases of exposure. Consequently, the actual prevalence of CO poisoning remains uncertain, and many cases likely go undetected Harper, 2004.

When CO enters the bloodstream, it binds to haemoglobin with a higher affinity than oxygen, displacing oxygen molecules Roy et al., 2024. This process reduces the amount of oxygen that haemoglobin can carry to vital organs throughout the body. Figure 6 shows how this problem is worsened

by environmental levels of carbon monoxide, which make it harder for the body to provide oxygen to vital organs like the heart and brain.

Indoor environments, especially those lacking proper ventilation, can pose a significant risk for carbon monoxide poisoning. The widespread adoption of natural gas heating systems, coupled with the increase in GHG emissions, has heightened the potential for such incidents in residential settings. In homes without adequate ventilation, individuals may unknowingly expose themselves to dangerous levels of carbon monoxide, often without realizing the danger until symptoms appear Vardoulakis et al., 2015.

Another source of concern is the use of indoor grills and smokers near or inside homes. Despite the belief that these devices are safe to use indoors, they can produce substantial amounts of carbon monoxide, which, even at low concentrations, can cause symptoms and pose health risks over time Marques et al., 2021. The binding of carbon monoxide to haemoglobin reduces oxygen transport to critical organs, posing significant health risks, particularly in poorly ventilated indoor spaces. Awareness of sources of carbon monoxide and its potential health impacts is crucial for preventing poisoning incidents in residential settings.

Every year, carbon monoxide poisoning leads to more than 20,000 visits to emergency departments across the United States, making it a significant public health concern. Carbon monoxide poisoning can present with a diverse range of symptoms, some of which are less common but potentially severe. These may include heart-related issues like myocardial ischemia, which refers to reduced blood flow to the heart muscle, and atrial fibrillation, an irregular heartbeat. Respiratory complications such as pneumonia and pulmonary edema, where fluid accumulates in the lungs, are also possible. Metabolic disturbances like hyperglycaemia, elevated blood sugar levels, and lactic acidosis, a buildup of lactic acid in the body, may occur as well. Additionally, carbon monoxide poisoning can lead to muscle damage, known as muscle necrosis, and acute kidney failure. Skin abnormalities, including various lesions, and impairments in sensory functions are other potential manifestations. Each of these symptoms underscores the broad and serious impact that carbon monoxide poisoning can have on multiple body systems Waite et al., 2014. Furthermore, prolonged exposure

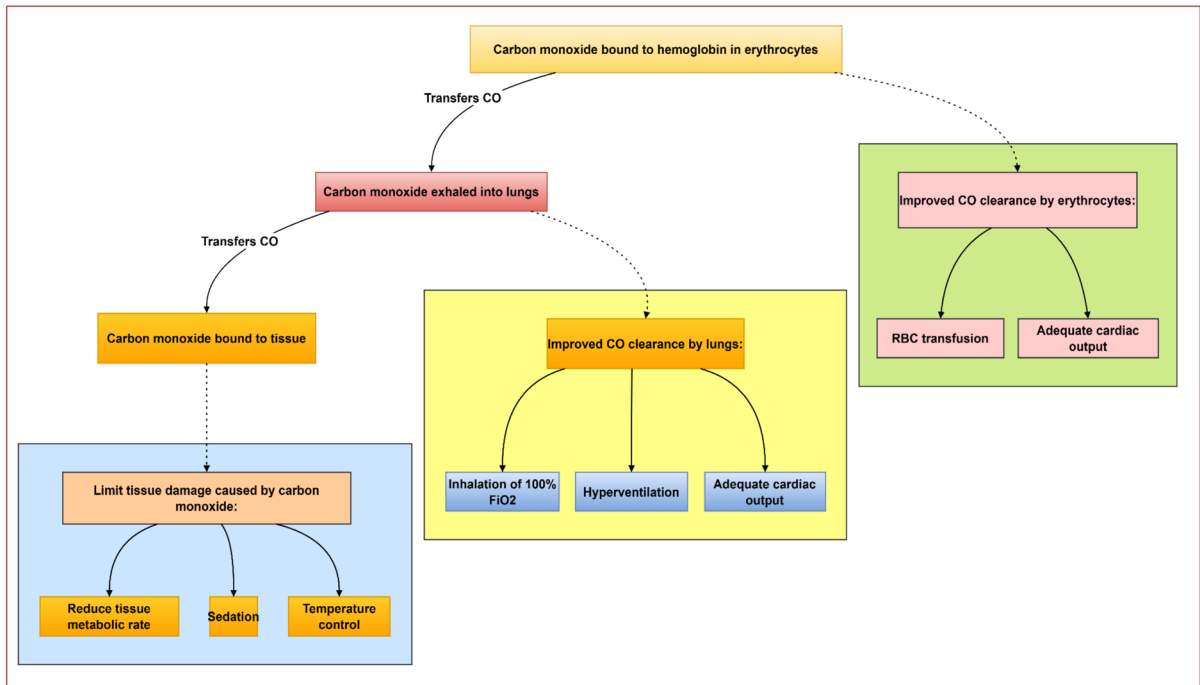


Fig. 6 Impact of CO in human body

to carbon monoxide can cause lasting damage to the heart, significantly impacting overall life expectancy Lee et al., 2015.

While outdoor concentrations of carbon monoxide are typically low and not a major concern, elevated levels in outdoor environments can pose a considerable risk, particularly to individuals with pre-existing heart conditions. Such individuals are especially vulnerable during situations that increase their heart's oxygen demand, such as periods of stress or physical exertion. This heightened susceptibility can result in symptoms like angina (chest pain) and reduced oxygenation of cardiac tissues Chu et al., 2021.

Despite its significant health impacts, the global environmental effects of carbon monoxide are generally considered minimal. However, locally near its sources of emission, carbon monoxide can interact with other air pollutants, potentially exacerbating air quality issues. During the summer months, carbon monoxide may oxidize to form carbon dioxide, a potent GHG contributing to climate change, which is harmful to human health and ecosystems (Fig. 6). In summary, while carbon monoxide primarily poses a direct health risk through poisoning,

its environmental interactions near emission sources warrant attention for localized air quality management and understanding its role in broader atmospheric processes.

4.1.3 Sulphur Dioxide

Sulphur dioxide (SO₂) is a highly detrimental gas classified within the broader group of sulphur oxides, predominantly released into the atmosphere during the combustion of fossil fuels or materials containing sulphur. This chemical poses significant risks to both environmental integrity and human health. When emitted, SO₂ has been documented to cause extensive damage to vegetation, including trees and plants, by inhibiting their growth and disrupting fragile ecosystems. Moreover, it contaminates water bodies, further compounding ecological stress.

Beyond its environmental impact, sulphur dioxide is a potent contributor to human health problems. It has been associated with exacerbating pre-existing conditions of the heart and lungs, while also triggering respiratory ailments in exposed populations. Acid rain, a direct consequence of SO₂ emissions, poses

additional risks to a wide array of natural resources such as trees, plants, sensitive ecosystems, and freshwater streams, hampering the growth and health of vegetation Zhang et al., 2023.

Furthermore, sulphur dioxide plays a pivotal role in atmospheric phenomena by contributing to the formation of dense fog and haze. This atmospheric condition not only reduces visibility but also complicates breathing, particularly affecting vulnerable demographics Javed et al., 2021.

Scientific studies have provided insight into the biological effects of SO₂ exposure. Experimental evidence indicates that activated sulphur dioxide can increase micronuclei in plant species of the genus *Tradescantia*. In controlled settings, mammalian eggs (such as those from cows and ewes) exposed to SO₂ in vitro, where metabolic activity is absent, have exhibited increased chromosome aberrations Wasielek-Politowska & Kordowitzki, 2022. Moreover, correlations have been established between sulphur dioxide exposure and heightened micronuclei in plants, as well as genetic alterations observed in *Saccharomyces cerevisiae*, a model organism in genetic research as shown in Fig. 7. Limited but suggestive evidence also suggests that occupational exposure to SO₂ through inhalation may induce clastogenic effects.

Despite significant research efforts, numerous unanswered questions persist regarding the precise impacts of sulphur dioxide (SO₂) and other sulphur oxides on human health. The complexity arises partly

due to the occasional presence of sulphur dioxide in particulate form and its interactions with high humidity levels, complicating the differentiation of its effects from other pollutants Khalaf et al., 2022. Outdoor activities, especially for individuals with asthma, can lead to transient breathing difficulties when exposed to elevated concentrations of sulphur dioxide Reno et al., 2015.

Individuals who suffer from asthma may experience a decrease in lung function and symptoms such as wheezing, chest tightness, or shortness of breath when they are exposed to elevated levels of certain gases during moderate physical activity. This brief exposure can significantly impact their respiratory health, highlighting the vulnerability of asthmatics to environmental pollutants.

"Acid rain," a term encompassing precipitation with high concentrations of nitric and sulfuric acids, poses substantial environmental and societal risks. These acids originate primarily from the combustion of fossil fuels, which releases nitrogen oxides and sulfur oxides into the atmosphere. Upon release, these compounds can undergo transformations in the atmosphere and eventually return to the Earth's surface as wet precipitation or dry deposition.

The transport of these acidic substances over long distances by wind means that their effects can extend far beyond their original emission sources Wikel, 2022. When acid rain falls, it accelerates the process of acidification in soils and bodies of water. This acidification is harmful to plants and aquatic life,

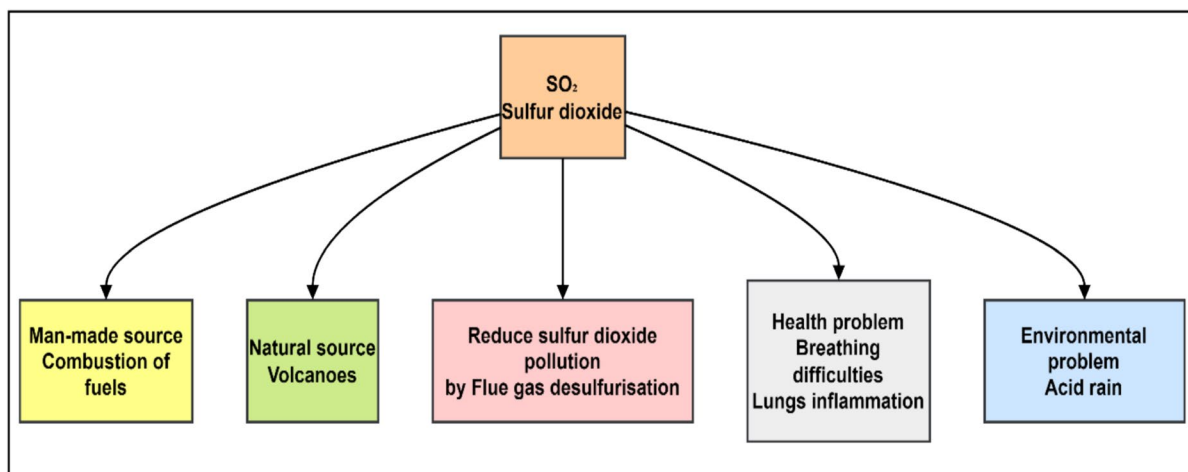


Fig. 7 Impact of Sulphur dioxide

making water bodies unsuitable habitats for certain species. Furthermore, the acidic nature of the rain accelerates the deterioration of important cultural and historical structures, including monuments, buildings, and sculptures.

The state of Massachusetts has experienced significant environmental damage to its lakes, ponds, rivers, and soil due to acid rain, affecting local ecosystems and the organisms that depend on them Saxena et al., 2024c. The primary contributors to acid rain, sulphur dioxide and nitrogen oxides, not only increase acidity levels in natural environments but also contribute to the formation of fine particulate matter that poses health risks to humans. Sulphur dioxide, in particular, plays a critical role in the formation of these harmful particles.

Ecosystems worldwide are adversely affected by acid rain, with significant impacts observed in forests and urban areas near industrial emissions. Urban vegetation, for instance, can be particularly susceptible to damage from sulphur dioxide deposition Englande et al., 2015. Lichens, known for their sensitivity to environmental pollutants, serve as bio-indicators of air quality and struggle to thrive in areas with elevated sulphur dioxide levels.

Efforts to reduce sulphur dioxide emissions not only mitigate the environmental and health impacts associated with acid rain but also yield secondary benefits. By decreasing sulphur dioxide emissions, strategies aimed at improving air quality can also reduce the formation of fine sulphate particles and other harmful pollutants Cui et al., 2019, Jerin et al., 2022. These measures are crucial for safeguarding both human health and the integrity of natural and cultural heritage sites affected by acid rain.

4.1.4 Nitrogen Oxides

Nitrogen dioxide (NO₂) is not commonly released into the atmosphere in its primary form. Instead, it is produced through chemical interactions involving nitrogen oxide (NO) and various other airborne compounds. A significant portion of NO₂ in the environment originates from human activities, particularly the burning of fossil fuels such as coal, oil, and gasoline, with vehicular emissions being a notable contributor Sharma et al., 2024 and Saxena et al., 2024d. In addition to transportation, several industrial processes are responsible for NO₂ emissions. These

processes include the production of nitric acid, which is extensively used in fertilizers and explosives, as well as welding activities, metal refining, and oil refining operations. Other sources include commercial manufacturing practices and food processing activities Lalah et al., 2022. Through these processes, industries significantly contribute to the overall levels of nitrogen dioxide in the atmosphere, impacting air quality and public health.

Natural sources like bacteria and volcanic activity also release nitrogen oxides into the environment in significant quantities. The control of combustion processes plays a crucial role in managing NO_x emissions. Strategies involve adjusting combustion parameters to lower peak temperatures, reducing the time spent at high temperatures, and optimizing oxygen levels in combustion zones Abdel-Shafy & Mansour, 2018. It's noteworthy that NO_x production increases exponentially at temperatures exceeding 1200 degrees Celsius.

Nitrogen dioxide primarily impacts human health through its effects on the respiratory system. Children are particularly vulnerable, with exposure to NO₂ linked to increased susceptibility to respiratory infections and potential long-term impacts on lung function. Elevated NO₂ levels in ambient air are associated with higher rates of hospitalizations and deaths related to respiratory disorders. Moreover, NO₂ can compromise the lungs' natural defenses against bacteria, potentially worsening asthma symptoms and respiratory conditions Cunillera-Montcusí et al., 2022.

Table 3 categorizes different types of nitrogen oxides and highlights their various industrial and environmental applications. This classification helps in understanding the diverse sources and uses of nitrogen oxides, underscoring their significance in both natural and industrial settings.

4.1.5 Polycyclic Aromatic Hydrocarbons

Carbohydrates, crude oil, and petrol are examples of compounds that include polycyclic aromatic hydrocarbons (PAHs), a group of chemicals characterized by multiple fused aromatic rings. PAHs are produced during the incomplete combustion of organic materials such as coal, tobacco, oil, and gas. This process can lead to the direct formation of PAHs or their attachment to airborne particles, which can be inhaled or deposited on surfaces. Additionally, high-temperature cooking methods, like grilling or

Table 3 Uses of NO₂ and NO

Use	Nitrogen dioxide (NO ₂)	Nitric oxide (NO)
Catalyst in the production of nitric acid for fertilizers	✓Cunillera-Montcusí et al., 2022	
Bleaching agent for flour	✓Cunillera-Montcusí et al., 2022	
Nitrating agent for chemical explosives	✓Cunillera-Montcusí et al., 2022	
Intermediary in nitric acid production	✓Cunillera-Montcusí et al., 2022	
Room temperature sterilizing agent	✓Cunillera-Montcusí et al., 2022	
Polymerization inhibitor for acrylates	✓Cunillera-Montcusí et al., 2022	
Oxidizer in rocket fuel	✓Cunillera-Montcusí et al., 2022	
Lowers blood pressure and improves blood flow		✓Dorantes-Aranda, 2023
Potential medical uses in supplements for specific disorders		✓Dorantes-Aranda, 2023
Naturally produced in the human body, crucial for blood vessel healing		✓Dorantes-Aranda, 2023

barbecuing meat, can generate significant amounts of PAHs, contributing to their presence in the environment and in food.

Naphthalene, a specific synthetic PAH, is commonly utilized in the United States for manufacturing mothballs and various other chemical products due to its properties. The burning of tobacco in cigarettes is another major source of PAHs, releasing numerous types into the air, which can pose health risks to smokers and those exposed to second-hand smoke. Therefore, PAHs are prevalent in many everyday scenarios, from industrial activities to household practices, highlighting the widespread nature of these potentially hazardous compounds.

In rural regions, the major contributors to PAH pollution stem from domestic and agricultural activities. These include practices like burning crop residues, using solid fuels for cooking and heating, and applying chemical treatments to crops. Conversely, in urban environments, PAH contamination primarily originates from industrial processes, vehicular emissions, and various residential sources. The dense network of transportation, extensive industrial operations, and routine domestic activities like cooking and heating significantly add to the urban PAH load.

PAH concentrations exhibit notable seasonal fluctuations, reaching their peak during the winter months. This is followed by a gradual decrease through spring and autumn, with the lowest levels observed in summer. Several factors contribute to this seasonal pattern. During winter, the incomplete combustion of fossil fuels is more prevalent due to heightened heating needs, which leads to higher emissions of PAHs. Additionally, the colder temperatures and

reduced sunlight in winter hinder the photodegradation of PAHs, causing these pollutants to persist longer in the environment. Poor atmospheric diffusion conditions, such as calm winds and low temperatures, further exacerbate the accumulation of PAHs during the colder months, as these meteorological conditions limit the dispersion and dilution of airborne pollutants Georgaki et al., 2023.

PAHs tend to accumulate in soil and sediment, which can result in the contamination of groundwater, plants, and the food chain. When PAHs are present in the environment, they can seep into the soil and persist for long periods due to their chemical stability. This persistence leads to significant environmental contamination. For example, PAHs in contaminated soil can be taken up by plant roots and then transported to other parts of the plant, including leaves, stems, and fruits. This process means that PAHs can enter the food chain when humans and animals consume these plants. The widespread presence of PAHs in the environment makes exposure to these harmful compounds almost unavoidable. Individuals can be exposed to PAHs through three main pathways: inhalation, ingestion, and skin contact. Inhalation can occur when PAHs are present in the air, particularly in areas with significant industrial emissions or where combustion processes, such as vehicle exhaust or tobacco smoke, release these compounds. Ingestion happens when PAHs enter the body through consuming contaminated food or water. Finally, skin contact with contaminated soil or water can also lead to PAH absorption Steinmaus, 2016.

To minimize cancer risk, it is crucial to consider a no-effect threshold when assessing the health risks of

PAHs. This threshold helps determine safe exposure levels, ensuring public health protection by identifying the amount of PAH exposure that does not significantly increase cancer risk. This approach is particularly important for certain PAHs, which are known to be carcinogenic air pollutants Howard et al., 2021. According to the WHO, existing guidelines for BaP, a specific type of PAH, may not provide sufficient protection against health issues other than cancer. Consequently, ongoing research is necessary to explore the health impacts of PAH exposure beyond cancer, including their effects on other health endpoints.

In aquatic and marine environments, PAHs can have detrimental effects on cardiovascular health, impacting all life stages of fish. The heart, being a vital organ, plays a critical role in extracting oxygen from the environment and delivering it to the mitochondria within cells. Damage to the heart caused by PAH exposure can therefore have far-reaching consequences for aquatic life Schwartz et al., 2021.

Furthermore, PAHs are not limited to environmental pollution but are also found in various categories of food and beverages. This widespread presence makes PAHs a significant concern for both environmental health and food safety Seo et al., 2023. The comprehensive study of PAHs and their impacts remains crucial to developing effective guidelines and protective measures against their harmful effects.

PAHs represent a complex compound consisting of both lighter and heavier molecules. Among these, BaA, Chr, BbF, and BaP are identified as the principal PAHs found in foods according to the European Commission Omarova et al., 2018. These compounds can contaminate food through various pathways, including production processes and environmental pollution.

PAHs originate from multiple sources such as the production and utilization of coal, oil, and petroleum products, as well as the combustion of wood and other plant materials. Soil contamination can occur when these substances are burned, leading to PAHs being present in agricultural lands where food is grown Wang & Qian, 2021. Furthermore, intentional industrial processes like the cracking of complex hydrocarbons in petroleum can produce what are known as pyrogenic PAHs. Urban areas, due to higher concentrations of industrial activities, often experience elevated levels of these PAHs Koutsoyianis, 2021.

The impact of PAHs extends beyond their chemical structure. They are known to absorb visible and UVA light and can generate reactive species and intermediates. These reactive compounds have the potential to cause harm to biological structures such as cell membranes, nucleic acids (DNA and RNA), and proteins Skendžić et al., 2021.

Table 4 provides a comparative assessment of air pollutants, focusing on CO, SO₂, and PAHs. It highlights their sources, key reactions, health and environmental effects, and strategies for mitigation, emphasizing the complexity of addressing diverse pollution challenges.

Research efforts have been dedicated to understanding human exposure to PAHs through diet and their overall environmental contribution Sandifer et al., 2015. This underscores the importance of monitoring and controlling PAH levels in food production and processing to mitigate potential health risks associated with these ubiquitous environmental pollutants.

4.2 Air Pollution and Human Health

Exposure to elevated levels of air pollution poses numerous adverse consequences for physical health. Individuals exposed to such pollution face increased risks of developing respiratory infections, heart problems, and lung cancer Sheoran et al., 2022. These health impacts can manifest both in the short term, causing immediate discomfort and illness, and in the long term, contributing to chronic conditions and reduced life expectancy. Moreover, individuals who already suffer from pre-existing health conditions are particularly susceptible to exacerbated symptoms and complications due to air pollution.

Children, whose respiratory systems are still developing, are at risk of long-term respiratory issues. The elderly, whose immune systems may be compromised, are more likely to experience severe health consequences from pollution exposure. Additionally, socioeconomically disadvantaged populations, including those with limited access to healthcare and living in polluted urban areas, face heightened risks Sheoran et al., 2022.

Among the pollutants, PM_{2.5} particles are especially harmful to human health. These fine particles are small enough to penetrate deep into the lungs, leading to respiratory and cardiovascular problems.

Table 4 Comparative assessment of air pollutants: CO, SO₂, and PAH

Pollutant	Sources	Key Reactions and Transformations	Effects	Long-term Environmental Impact	Mitigation Strategies	Ref
CO	Incomplete combustion of fossil fuels (vehicles, wildfires, industrial processes)	Oxidized in the atmosphere to form CO ₂	Health: Binds with hemoglobin, causing hypoxia and acute poisoning	No direct global warming effect but contributes to ozone production	Catalytic converters for vehicles	Sheoran et al., 2022
	Photochemical reactions with hydrocarbons	Intermediate in the formation of ground-level ozone in urban environments	Environmental: Amplifies VOCs and NO _x in smog formation	Regional air quality degradation affecting ecosystems	Transition to cleaner fuels	Liu et al., 2022
SO ₂	Combustion of sulphur-containing fuels (coal, oil, natural gas)	Forms sulphuric acid in the presence of moisture, leading to acid rain	Health: Irritates respiratory pathways; long-term exposure linked to cardiovascular issues	Acid rain damages aquatic systems, soils, and infrastructure	Public campaigns for CO detector installations	Harper, 2004
	Volcanic emissions	Converts to sulphate aerosols	Environmental: Acidification of water bodies and soil	Sulphate aerosols reflect sunlight but exacerbate regional pollution	Implement desulphurization technologies	Zhang et al., 2023
	Metal smelting and industrial processes				Regulatory caps on SO ₂ emissions	Javed et al., 2021
PAHs	Incomplete combustion of carbon-based materials (coal, wood, petroleum products)	Atmospheric PAHs can bind to particulates, increasing toxicity	Health: Carcinogenic; exposure through inhalation and contaminated food/water	Long-term soil acidification reduces agricultural productivity	Promote renewable energy alternatives	Khalaf et al., 2022
	Vehicle exhaust and industrial processes	Degraded by photochemical reactions or deposited onto soil and water	Environmental: Bioaccumulation disrupts terrestrial and aquatic ecosystems	Persistent in the environment due to low degradability	Enhance industrial emissions controls (filters, scrubbers)	Georgaki et al., 2023
	Natural sources: forest fires			Biodiversity loss and contamination of sediments in water systems	Stricter waste management regulations	Steinmaus, 2016
					Foster research on bioremediation of contaminated sites	Wang & Qian, 2021, Skendžić et al., 2021, Saxena et al., n.d

They have been directly linked to increased mortality rates, particularly affecting vulnerable populations Maciejczyk et al., 2021. The variability in health impacts from air pollution depends on multiple factors, such as age, geographical location, overall health status, and lifestyle choices.

Factors such as residential proximity to industrial facilities, traffic congestion, and inadequate environmental regulations contribute to higher pollution levels in these communities Robb & Techtman, 1981. Consequently, these populations are more likely to suffer from the negative health effects associated with air pollution, perpetuating health disparities Sellami and Elleuch, 2024.

Regular exposure to wood smoke presents another significant health concern. Wood smoke contains a complex mixture of pollutants that pose significant risks to respiratory health. Among these pollutants are fine particles, which can penetrate deep into the lungs and cause various respiratory issues. VOCs are also present in wood smoke, and they contribute to air pollution and can irritate the respiratory system. Additionally, wood smoke contains CO, a toxic gas that can interfere with the oxygen-carrying capacity of blood, leading to potential respiratory and cardiovascular problems. Dioxins, another component of wood smoke, are highly toxic compounds that can have long-term health effects, including respiratory issues and an increased risk of cancer. The combination of these pollutants in wood smoke can have severe adverse effects on respiratory health, making it a significant environmental and public health concern Marques et al., 2021. This underscores the importance of addressing not only industrial and vehicular sources of air pollution but also household sources like wood burning to mitigate health risks.

Wood smoke, beyond just containing soot, gases, and wood tars, encompasses a diverse array of other substances. The short-term exposure to fine particulate matter present in the air due to wood smoke can notably exacerbate existing lung conditions, provoke asthma attacks, and lead to episodes of acute bronchitis. Furthermore, this exposure can potentially elevate the risk of respiratory infections, posing significant health risks to those affected. Research has also identified a link between brief exposures to these particles and occurrences of irregular heartbeats and heart attacks during clinical trials Ishika (2023). Furthermore, these fine particles are associated with the

development of several serious diseases, including COPD, chronic bronchitis, cardiovascular disease, and there is even evidence suggesting a possible link to lung cancer.

Ozone, normally beneficial high in the atmosphere where it protects us from harmful ultraviolet radiation, becomes problematic at ground level as a component of smog. This ground-level ozone forms when oxygen oxides and volatile organic compounds, both products of fossil fuel combustion, react in the presence of sunlight Zhang et al., 2023. Exposure to ozone, even for a short duration, can result in a variety of acute symptoms including chest pain, persistent coughing, and throat irritation. These immediate health effects can be quite discomforting and indicative of the body's adverse reaction to the pollutant. When exposure extends over a longer period, the health implications become more serious and enduring. Long-term exposure to ozone is associated with the gradual deterioration of lung function. This decline can manifest in reduced breathing capacity and overall respiratory efficiency. Furthermore, there is a significant correlation between prolonged ozone exposure and the heightened risk of developing COPD.

In addition to these risks, continuous exposure to ozone can exacerbate pre-existing lung conditions. Individuals with respiratory ailments such as asthma, bronchitis, and emphysema may experience a worsening of their symptoms. Over time, the severity of these conditions can increase, leading to more frequent and intense episodes of respiratory distress. This cumulative effect highlights the insidious nature of ozone as a persistent environmental hazard, capable of inflicting long-lasting damage to respiratory health Gautam et al., 2023.

The combustion of sulphur-containing fossil fuels results in the emission of sulphur dioxide gas into the atmosphere Naidu et al., 2021. This substance aggravates asthma symptoms, irritates the eyes, and can cause respiratory infections due to its negative impact on the circulatory system. Furthermore, sulphur dioxide is an essential component of acid rain, which is responsible for the devastation of ecosystems worldwide due to sulphuric acid, a result of sulphur dioxide's reaction with atmospheric water.

Collaborative efforts among experts from various disciplines, including epidemiology, human observation, and toxicology, are underway to comprehensively assess the impact of air pollution on

vulnerable populations Wang et al., 2022. Their primary objective is to devise effective strategies aimed at safeguarding public health, with a particular focus on individuals. To achieve this, it is crucial to comprehend the entire range of health impacts that result from both short-term and long-term exposure to smoke generated by wildfires and controlled burns. These types of fires, collectively known as "wildland fires," pose significant risks. Therefore, a thorough understanding of the health effects associated with wildland fire smoke is essential for accurately assessing these risks and implementing appropriate mitigation measures Pingali, 2012.

4.3 Air Pollution's Effects on the Environment

Air pollution stands as a formidable challenge in contemporary society Siddiqua et al., 2022. Its impact extends across multiple dimensions, contributing significantly to climate change and profoundly affecting public and individual health. The consequences are stark, with elevated rates of disease and mortality directly linked to exposure to pollutants Srivastava et al., 2023. Furthermore, climate change exacerbates these health risks by prolonging pollen seasons and fostering conditions favourable for the proliferation of allergenic air pollutants such as mold and pollen, driven by more frequent and intense weather events and increased flooding Ewuzie et al., 2021.

The crucial impact of GHG, especially CO_2 and CH_4 , is central to the climate change problem. These gases are crucial in modifying the dynamics of Earth's climate. The role of these GHG in driving

climate change underscores the urgency of addressing emissions to mitigate their far-reaching consequences Srivastava et al., 2024. Notably, in 2018, the United States' GHG emissions predominantly comprised carbon dioxide (81%) and CH_4 (10%) Srivastava & Roy, 2023.

Figure 8 illustrates how air pollution affects the environment, including climate change, ecosystem degradation, and health impacts on flora and fauna. It emphasizes the interconnected nature of these effects, underscoring the urgency of mitigation strategies.

Recent studies Srivastava & Roy, 2023 highlight the significant environmental impacts of oil and gas drilling, particularly in the context of methane emissions. These pollutants not only alter the chemical composition of these environments but also lead to the release of harmful metals and the depletion of essential nutrients in freshwater and forest soils. These changes have detrimental effects: low pH levels in freshwater can lead to the death of fish populations, while in forest soils, nutrient loss can result in forest degradation.

In recent decades, efforts to reduce sulphur dioxide emissions have been successful in mitigating acidification issues. However, ground-level ozone remains a significant threat to vegetation. Ozone, which penetrates plant leaves, disrupts the process of photosynthesis, thereby hindering plant growth. Widespread changes to ecosystem structure, as well as decreases in habitat quality and biodiversity, have been associated to elevated amounts of ozone in the ground. In agricultural settings, ozone pollution can reduce crop yields and impair forest growth.

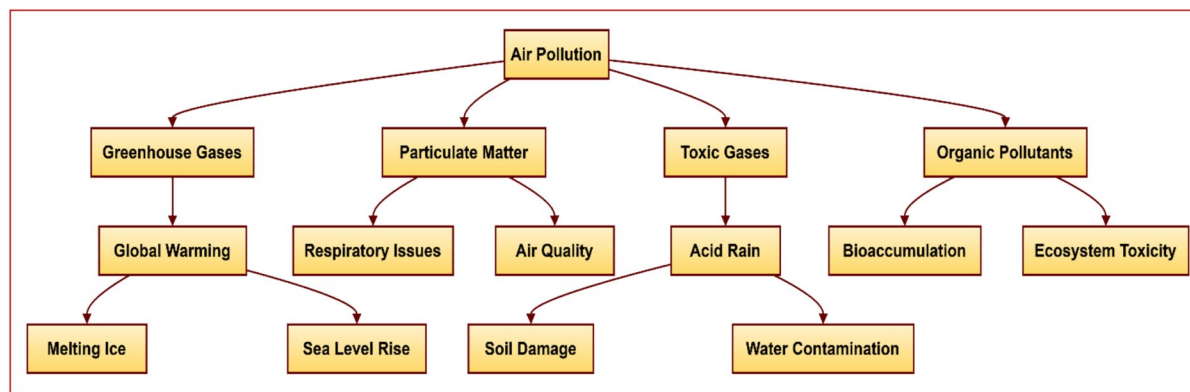


Fig. 8 Impact of air pollution on the environment

In terrestrial ecosystems, nitrogen deposition from the atmosphere can lead to eutrophication under certain conditions. When nitrogen deposition surpasses critical loads, it can trigger dangerous levels of eutrophication, affecting ecosystem health and biodiversity. The concept of "critical load" emphasizes that exceeding certain pollution thresholds can disrupt both terrestrial and aquatic ecosystems, altering the composition and diversity of species. Conversely, keeping pollution levels below these critical thresholds allows ecosystems to absorb deposited pollutants without significant disruption.

Particulate matter (PM) pollution, including PM10 and PM2.5, also poses serious health risks. Coarse particles can affect the nasal and upper respiratory systems, while fine particles, due to their smaller size, can penetrate deeper into the lungs and even enter the bloodstream. Also, research shows that kids whose exposure to PM2.5 levels are higher may have problems with brain development.

Overall, understanding and mitigating the environmental and health impacts of these pollutants are crucial for sustainable development and public health protection. Continuing research and effective policies are essential to address these complex challenges and minimize their detrimental effects on ecosystems and human well-being.

Table 5 highlights the direct correlation between pollutant exposure and its impact on human health, emphasizing vulnerable groups at higher risk.

5 Conclusion

The challenges posed by human-induced pollution and climate change demand urgent, multifaceted action. Below are the key takeaways and implications for addressing these issues effectively:

- **Human Contribution to Pollution:**

Human activities are the primary drivers of water and air pollution, contributing significantly to climate change.

Solutions require collective responsibility and comprehensive strategies across societal and economic systems.

- **Environmental Preservation and Restoration:**

Immediate Action: Investments are needed to rectify ecological damage caused by pollution.

Sustainable Practices: Adoption of emission-reducing technologies, conservation of natural resources, and biodiversity promotion are critical.

Restoration Initiatives: Beyond preventing degradation, proactive restoration of habitats ensures thriving ecosystems.

- **Educational and Technological Advancements:**

Public awareness campaigns can shift behaviors like littering and deforestation.

Advanced technologies targeting industrial and power plant emissions are crucial for mitigating air pollution and aligning with global environmental objectives.

- **Policy and Governance:**

Governments must create evidence-based policies that address the broad impacts of pollution on public health and ecosystems.

Local frameworks tailored to specific environmental conditions can serve as models for international adoption.

- **Role of Research and Innovation:**

Ongoing research is vital for discovering innovative pollution mitigation strategies.

Integrating research findings into policy ensures adaptive and forward-looking environmental health strategies.

- **Global Implications:**

Efforts at local and regional levels collectively contribute to global sustainability.

Coordinated international action, inspired by successful local practices, is imperative for managing climate change and pollution at scale.

Table 5 Summarizing the relationship between various air pollutants and their health impacts

Pollutant	Source	Short-Term Health Effects	Short-Term Numeric Impact	Long-Term Health Effects	Long-Term Numeric Impact	Most Affected Groups
Particulate Matter (PM2.5, PM10)	Combustion (vehicles, industrial processes), wildfires	Increases respiratory hospitalizations, aggravated asthma	10% increase per 10 µg/m ³ PM2.5 levels	Cardiovascular mortality, lung cancer risk	7% increase (mortality), 15% increase (lung cancer risk)	Children, elderly, individuals with respiratory conditions
NOx	Vehicle emissions, power plants, industrial activities	Irritation of airways, reduced lung function	20–30% decrease in lung function tests in children	Chronic asthma development in children	13% higher risk	Urban populations, individuals with pre-existing lung conditions
SO2	Burning of fossil fuels (coal, oil)	Wheezing, asthma-related ER visits	5–10% increase in asthma-related visits	Chronic bronchitis, heart disease	30% increase (chronic bronchitis)	People with asthma, children, elderly
O3	Reaction of sunlight with NOx and VOCs	Reduced lung capacity, throat irritation	10–20% reduction in lung capacity on high-ozone days	Asthma development, lung damage	12% increase in asthma risk	Outdoor workers, children, individuals with asthma
CO	Incomplete combustion of fossil fuels	Dizziness, cognitive impairment	High-level exposure > 30 ppm; low-level affects cognition	Cardiovascular events	15% increase in urban areas	People with heart conditions, pregnant women
VOCs	Paints, solvents, vehicle emissions	Eye irritation, headaches	30–50% affected by exposure	Liver disease, carcinogenic effects	50% increase in liver disease risk	Industrial workers, urban populations
Lead	Industrial emissions, leaded fuels, contaminated water	Developmental delays in children	25–50% affected at > 10 µg/dL blood lead levels	Cardiovascular and kidney diseases	3 × higher risk	Children, pregnant women
NH3	Agricultural activities, waste management	Respiratory tract irritation	60% of agricultural workers affected	Aggravated asthma symptoms	35% affected by long-term exposure	Agricultural workers, individuals with asthma

By uniting these efforts—ranging from education to policy reform and technological innovation—societies can address the pressing challenges of environmental degradation and climate change, ensuring a healthier planet for current and future generations.

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Declarations

Ethical approval All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for Authors and are aware that with minor exceptions, no changes can be made to authorship once the paper is submitted.

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References

- Abdel-Shafy, H. I., & Mansour, M. S. M. (2018). Solid waste issue: Sources, composition, disposal, recycling, and valorization. *Egyptian Journal of Petroleum*, 27, 1275–1290.
- Adamo, N., Al-Ansari, N., & Sissakian, V. (2021). Review of climate change impacts on human environment: Past, present and future projections. *Engineering*, 13, 605–630.
- Adams, G. O., Ogedegbe, P. E., & Tawari-Fufeyin, P. (2016). Assessment of presence of heavy metals and other pollution burden parameters and their effect on water quality in Benin City. *Environmental Quality Management*, 26, 65–87.
- Adla K, Dejan K, Neira D, Dragana Š. Degradation of ecosystems and loss of ecosystem services. *One Heal*. 2022;281–327.
- Agbasi JC, Egbueri JC, Ayejoto DA, Unigwe CO, Omeka ME, Nwazelibe VE, Ighalo JO, Pande CB, Fakoya AA. The Impact of Seasonal Changes on the Trends of Physico-chemical, Heavy Metal and Microbial Loads in Water Resources of Southeastern Nigeria: A Critical Review. 2023;505–539.
- Alao, J. O., Fahad, A., Abdo, H. G., Ayejoto, D. A., Almo-hamad, H., Ahmad, M. S., Nur, M. S., Danjuma, T. T., Yusuf, M. A., Francis, O. T., & Joy, A. O. (2023). Effects of dumpsite leachate plumes on surface and groundwater and the possible public health risks. *Science of the Total Environment*, 897, 165469.
- Aliyu, A., & Amadu, L. (2017). Urbanization, cities, and health: The challenges to Nigeria – a review. *Annals of African Medicine*, 16, 149.
- Almusawi, M. B. H., Karim, A. T. B. A., & Ethaib, S. (2022). Evaluation of construction and demolition waste management in Kuwait. *Recycling*, 7, 88.
- Archer, E. J., Baker-Austin, C., Osborn, T. J., Jones, N. R., Martínez-Urtaza, J., Trinanes, J., Oliver, J. D., González, F. J. C., & Lake, I. R. (2023). Climate warming and increasing *Vibrio vulnificus* infections in North America. *Science and Reports*, 13, 3893.
- Arora, G., Sharma, T., Kashyap, T. K., Pant, P., Gupta, C., & Sharma, R. K. (2023). Rejuvenation and Restoration of Surface Water Quality Amid COVID-19 Lockdown: A Comprehensive Review in Indian Context. *Environmental Engineering Research*, 28(3), 220144.
- Arora K, Kumar P, Bose D, Li X, Kulshrestha S. Potential applications of algae in biochemical and bioenergy sector. 3 *Biotech*. 2021; 11:296.
- Ateia, M., Kanan, A., & Karanfil, T. (2020). Microplastics release precursors of chlorinated and brominated disinfection byproducts in water. *Chemosphere*, 251, 126452.
- Ayejoto, D. A., Egbueri, J. C., Enyigwe, M. T., Chiaghanam, O. I., & Ameh, P. D. (2022). Application of HMTL and novel IWQI models in rural groundwater quality assessment: A case study in Nigeria. *Toxin Reviews*, 41, 918–932.
- Ayejoto DA, Egbueri JC. Human health risk assessment of nitrate and heavy metals in urban groundwater in South-east Nigeria. *Acta Ecologica Sinica*2023.
- Ayejoto DA, Egbueri JC, Agbasi JC, Omeka ME, Unigwe CO, Nwazelibe VE, Ighalo JO, Pande CB. Influence of Seasonal Changes on the Quality of Water Resources in Southwestern Nigeria: A Review. 2023;423–447.
- Azanu D, Adu-Poku D, Saah SA, Appaw WO. Prevalence of Pharmaceuticals in Surface Water Samples in Ghana. *Journal of Chemistry*. 2021;1–11.
- Babcock-Jackson, L., Konovalova, T., Krogman, J. P., Bird, R., & Díaz, L. L. (2023). Sustainable fertilizers: Publication landscape on wastes as nutrient sources, wastewater treatment processes for nutrient recovery, biorefineries, and green ammonia synthesis. *Journal of Agriculture and Food Chemistry*, 71, 8265–8296.
- Bălă, G.-P., Răjnoveanu, R.-M., Tudorache, E., Motișan, R., & Oancea, C. (2021). Air pollution exposure—the (in)visible risk factor for respiratory diseases. *Environmental Science and Pollution Research*, 28, 19615–19628.
- Bashpa, P., Bijudas, K., Dileep, P., Mehar, S., Silviya, E., & Francis, T. (2024). Mechanical and thermal characterization of condom industry waste reinforced natural rubber composites: A circular economy approach. *Environmental Engineering Research*, 29(2), 230098.
- Bhaga, T. D., Dube, T., Shekede, M. D., & Shoko, C. (2020). Impacts of climate variability and drought on surface water resources in sub-Saharan Africa using remote sensing: A review. *Remote Sens.*, 12, 4184.
- Bhattarai, H., Tai, A. P. K., Val Martin, M., & Yung, D. H. Y. (2024). Impacts of changes in climate, land use, and

- emissions on global ozone air quality by mid-21st century following selected Shared Socioeconomic Pathways. *Science of the Total Environment*, 906, 167759.
- Bilung LM, Tahar AS, Yunus NE, Apun K, Lim YA-L, Nillian E, Hashim HF. Detection of cryptosporidium and cyclospora oocysts from environmental water for drinking and recreational activities in Sarawak, Malaysia. *BioMed Research International* 2017;1–9.
- Blanchet, C. C., Arzel, C., Davranche, A., Kahilainen, K. K., Secondi, J., Taipale, S., Lindberg, H., Loehr, J., Manninen-Johansen, S., Sundell, J., Maanan, M., & Nummi, P. (2022). Ecology and extent of freshwater browning - what we know and what should be studied next in the context of global change. *Science of the Total Environment*, 812, 152420.
- Bouchard, C., Dibernardo, A., Koffi, J., Wood, H., Leighton, P., & Lindsay, L. (2019). Increased risk of tick-borne diseases with climate and environmental changes. *Canada Communicable Disease Report*, 45, 83–89.
- Bradru P, Biswas A, Nair C, Sreevalsakumar S, Patil M, Kannampuzha S, Mukherjee AG, Wanjari UR, Renu K, Vellingiri B, Gopalakrishnan AV. Recent advances in green technology and Industrial Revolution 4.0 for a sustainable future. *Environmental Science and Pollution Research* 2022.
- Briffa J, Sinagra E, Blundell R. Heavy metal pollution in the environment and their toxicological effects on humans. *Heliyon*. 2020;6
- Brulé, T., Renán, X., & Colás-Marrufó, T. (2022). Potential impact of climate change on fish reproductive phenology: A case study in gonochoric and hermaphrodite commercially important species from the southern Gulf of Mexico. *Fishes.*, 7, 156.
- Chu EW, Karr JR. Environmental impact: concept, consequences, measurement. Ref. Modul. Life Sci. 2017.
- Chu L.M., Shaefi S., Byrne J.D., Alves de Souza R.W., Otterbein L.E. Carbon monoxide and a change of heart. *Redox Biol*. 2021; 48:102183.
- Crane-Kramer, G., & Buckberry, J. (2023). Changes in health with the rise of industry. *International Journal of Paleopathology*, 40, 99–102.
- Cui, Q., Huang, Y., Wang, H., & Fang, T. (2019). Diversity and abundance of bacterial pathogens in urban rivers impacted by domestic sewage. *Environmental Pollution*, 249, 24–35.
- Cunillera-Montcusí, D., Beklioğlu, M., Cañedo-Argüelles, M., Jeppesen, E., Ptacnik, R., Amorim, C. A., Arnott, S. E., Berger, S. A., Brucet, S., Dugan, H. A., Gerhard, M., Horváth, Z., Langenheder, S., Nejstgaard, J. C., Reinikainen, M., Striebel, M., Urrutia-Cordero, P., Vad, C. F., Zadereev, E., & Matias, M. (2022). Freshwater salinisation: A research agenda for a saltier world. *Trends in Ecology & Evolution*, 37, 440–453.
- D'Amato, G., Chong-Neto, H. J., & Monge Ortega, O. P. (2020). The effects of climate change on respiratory allergy and asthma induced by pollen and mold allergens. *Allergy*, 75, 2219–2228.
- D'Amato G, Holgate ST, Pawankar R, Ledford DK. Meteorological conditions, climate change, new emerging factors, and asthma and related allergic disorders. A statement of the World Allergy Organization. *World Allergy Organ. J*. 2015; 8:25.
- da Costa, J. P., Silva, A. L., Barcelò, D., Rocha-Santos, T., & Duarte, A. (2023). Threats to sustainability in face of post-pandemic scenarios and the war in Ukraine. *Science of the Total Environment*, 892, 164509.
- Dey, S., & Dhal, G. C. (2019). Materials progress in the control of CO and CO₂ emission at ambient conditions: An overview. *Materials Science for Energy Technologies*, 2, 607–623.
- Dorantes-Aranda, J. J. (2023). Harmful algae impacting aquatic organisms: Recent field and laboratory observations. *Toxins (Basel)*, 15, 339.
- Ebi, K. L., Vanos, J., Baldwin, J. W., Bell, J. E., Hondula, D. M., Errett, N. A., Hayes, K., Reid, C. E., Saha, S., Spector, J., & Berry, P. (2021). Extreme weather and climate change: Population health and health system implications. *Annual Review of Public Health*, 42, 293–315.
- Echendu, A. J., Okafor, H. F., & Iyiola, O. (2022). Air pollution, climate change and ecosystem health in the Niger Delta. *Sociology of the Sciences*, 11, 525.
- Eckelman MJ, Sherman J. Environmental impacts of the U.S. health care system and effects on public health. *PLoS One* 2016;11.
- Edo, G. I., Onoharigho, F. O., Emakpor, G. L., & Akpogheli, P. O. (2022). The physicochemical analysis and health benefits of fresh and branded honey produced in Delta State Nigeria. *Journal of Analytical & Pharmaceutical Research*, 11, 66–72.
- Edo, G. I., Onoharigho, F. O., Jikah, A. N., Ezekiel, G. O., Essagah, A. E. A., Ekokotu, H. A., Ugbune, U., Oghrora, E. E. A., Emakpor, O. L., Ainyanbor, I. E., Akpogheli, P. O., Ojulari, A. E., Okoronkwo, K. A., & Owheru, J. O. (2024a). Evaluation of the physicochemical, phytochemical and anti-bacterial potential of *Zingiber officinale* (ginger). *Food Chem. Adv.*, 4, 100625.
- Edo, G. I., Jikah, A. N., Onoharigho, F. O., Akpogheli, P. O., Agbo, J. J., Ekokotu, H. A., Ugbune, U., Ezekiel, G. O., Emakpor, O. L., Oghrora, E. E. A., Ojulari, A. E., Okoronkwo, K. A., Omoruwu, P. E., & Owheru, J. O. (2024b). The ameliorative effects of *Vernonia amygdalina* extract on superoxide dismutase and glutathione s-transferase on alloxan induced diabetes on male Wistar rats. *Food Chem. Adv.*, 4, 100620.
- Edo GI, Samuel PO, Nwachukwu SC. Bioactive Compounds and Biological Activities of Tiger Nut (*Cyperus esculentus* L.). 2023;1–28.
- Elsayed, A. E. A., Mohamed, M. L., Ahmed, F. S., & Attia, F. A. (2023). Using nano-magnesium oxide/bentonite composite for cadmium removal from industrial wastewater. *Environmental Engineering Research*, 28(2), 210545.
- Englande, A. J., Krenkel, P., & Shamas, J. (2015). *Wastewater treatment & water reclamation*. Elsevier.
- Ewuzie U, Aku NO, Nwankpa SU. An appraisal of data collection, analysis, and reporting adopted for water quality assessment: a case of Nigeria water quality research. *Heliyon*. 2021;7
- Ezugwu, C. K., Onwuka, O. S., Egbueri, J. C., Unigwe, C. O., & Ayejoto, D. A. (2019). Multi-criteria approach to water quality and health risk assessments in a rural agricultural province Southeast Nigeria. *Hydroresearch*, 2, 40–48.
- Falkenberg, L. J., Bellerby, R. G. J., Connell, S. D., Fleming, L. E., Maycock, B., Russell, B. D., Sullivan, F. J.,

- & Dupont, S. (2020). Ocean acidification and human health. *International Journal of Environmental Research and Public Health*, 17, 4563.
- Fallah Shayan, N., Mohabbati-Kalejahi, N., Alavi, S., & Zahed, M. A. (2022). Sustainable development goals (SDGs) as a framework for corporate social responsibility (CSR). *Sustainability*, 14, 1222.
- Fan, M., Zhang, P., Wang, C., & Sun, H. (2024). Highly efficient removal of ionic dyes in aqueous solutions using magnetic 3D reduced graphene oxide aerogel supported nano zero-valent iron. *Environmental Engineering Research*, 29(1), 230149.
- Faquihi, T. O., Landstra, E. N., van Hylckama Vlieg, A., et al. (2024). Per- and Polyfluoroalkyl Substances Concentrations are Associated with an Unfavorable Cardio-Metabolic Risk Profile: Findings from Two Population-Based Cohort Studies. *Expo Health*. <https://doi.org/10.1007/s12403-023-00622-4>
- Fernandes, J. P., Almeida, C. M. R., Salgado, M. A., Carvalho, M. F., & Mucha, A. P. (2021). Pharmaceutical compounds in aquatic environments—occurrence, fate and bioremediation prospective. *Toxics*, 9, 257.
- Florentina I, Io B. The effects of air pollutants on vegetation and the role of vegetation in reducing atmospheric pollution. Impact Air Pollut. Heal. Econ. Environ. Agric. Sources. InTech; 2011.
- Fu, J., Yao, Y., Huang, Z., et al. (2024). Prenatal Exposure to Neonicotinoids and the Associations with Neonatal Birth Outcomes and Maternal Health in South China. *Expo Health*. <https://doi.org/10.1007/s12403-024-00650-8A>
- Gautam, S. P., Silwal, A., Baral, B., Subedi, S., Lamichhane, N., Chapagain, N. P., & Adhikari, B. (2023). Influence of the rainfall and temperature oscillation on air quality in Kathmandu valley: The wavelet analysis. *Environmental Engineering Research*, 28(6), 220694.
- Gavrilescu, M. (2021). Water, soil, and plants interactions in a threatened environment. *Water*, 13, 2746.
- Georgaki, M.-N., Charalambous, M., Kazakis, N., Talias, M. A., Georgakis, C., Papamitsou, T., & Mytilaki, C. (2023). Chromium in water and carcinogenic human health risk. *Environments*, 10, 33.
- Ghorani-Azam, A., Riahi-Zanjani, B., & Balali-Mood, M. (2016). Effects of air pollution on human health and practical measures for prevention in Iran. *Journal of Research in Medical Sciences: The Official Journal of Isfahan University of Medical Sciences*, 21, 65.
- Gour, P., Kumar, J., Arland, S. E., Roy, L. D., & Rahman, N. (2024). Green synthesis of DL-homocysteine decorated magnetic nanoparticles for selective and efficient mercury remediation from simulated wastewater: Kinetics, isotherm, and mechanism studies. *Environmental Engineering Research*, 29(5), 230584.
- Grennfelt, P., Englerdyd, A., Forsius, M., Hov, Ø., Rodhe, H., & Cowling, E. (2020). Acid rain and air pollution: 50 years of progress in environmental science and policy. *Ambio*, 49, 849–864.
- Gu, Y., Man, J., Su, L., et al. (2023). Impact of Dietary Cadmium Intake from Rice on Renal Function Biomarker and Population Toxicokinetic Modelling. *Expo Health*. <https://doi.org/10.1007/s12403-023-00615-3>
- Harper, A. (2004). Carbon monoxide poisoning: Undetected by both patients and their doctors. *Age and Ageing*, 33, 105–109.
- Hayasaka, H., Yamazaki, K., & Naito, D. (2019). Weather conditions and warm air masses during active fire-periods in boreal forests. *Polar Science*, 22, 100472.
- Heo, J., Kim, P., Park, J., & Lee, J. (2025). Source identification of ultrafine particulate matter and ammonia emissions in the metropolitan area. *Environmental Engineering Research*, 30(1), 240074.
- Howard, I., Okpara, K., & Techato, K. (2021). Toxicity and risks assessment of polycyclic aromatic hydrocarbons in river bed sediments of an artisanal crude oil refining area in the Niger Delta Nigeria. *Water*, 13, 3295.
- Huggett, R. (2023). Soil as part of the Earth system. *Progress in Physical Geography: Earth and Environment*, 47, 454–466.
- Hussain, N. T., Jameel, A. A., & Murtadha, I. M. (2024). Sources, fate, effects, and analysis of microplastic in wastewater treatment plants: A review. *Environmental Engineering Research*, 29(1), 230040.
- Hwabejire HO, Akpogheli PO, Edo GI, Onoharigho FO, Agbo JJ. Microbiological properties, anti-nutritional and nutritional composition of spontaneously and starter culture fermented malted acha flour. Proceedings of the Indian National Science Academy 2023.
- Iruoghene Edo G. Analysis of phytochemical constituents and antioxidant potential of bitter Kola leaf extract towards bioactive food, nutrition and health resources. *International Journal of Medicinal Chemistry* 2022;11.
- Ishika, N. (2023). Development of an engineered face mask with optimized nanoparticle layering for filtration of air pollutants and viral pathogens. *Environmental Engineering Research*, 28(6), 230003.
- Jaishankar, M., Tseten, T., Anbalagan, N., Mathew, B. B., & Beeregowda, K. N. (2014). Toxicity, mechanism and health effects of some heavy metals. *Interdisciplinary Toxicology*, 7, 60–72.
- Javed A., Aamir F., Gohar U., Mukhtar H., Zia-UI-Haq M., Alotaibi M., Bin-Jumah M., Marc (Vlaic) R., Pop O. The potential impact of smog spell on humans' health amid COVID-19 rages. *International Journal of Environmental Research and Public Health* 2021; 18:11408.
- Jeong, J. E., Cho, I. A., & Lee, C. Y. (2024). Effect of organic additives on NO₂ removal in the simultaneous absorption of SO₂ and NO₂ into limestone slurry. *Environmental Engineering Research*, 29(4), 230513.
- Jerin DT, Sara HH, Radia MA, Hema PS, Hasan S, Urme SA, Audia C, Hasan MT, Quayyum Z. An overview of progress towards implementation of solid waste management policies in Dhaka, Bangladesh. *Heliyon*. 2022;8
- Jiang, J.-Q., Ashekuzzaman, S., Jiang, A., Sharifuzzaman, S., & Chowdhury, S. (2012). Arsenic contaminated groundwater and its treatment options in Bangladesh. *International Journal of Environmental Research and Public Health*, 10, 18–46.
- Kang, S., Ren, G., Lee, T., & Jo, Y. (2024). Correlation between carbonaceous materials and fine particulate matters in urban school classrooms. *Environmental Engineering Research*, 29(4), 230516.

- Khalaf E.M., Mohammadi M.J., Sulistiyani S., Ramírez-Coronel A.A., Kiani F., Jalil A.T., Almulla A.F., Asban P., Farhadi M., Derikondi M. Effects of sulfur dioxide inhalation on human health: a review. *Review Environmental Health* 2022.
- Kim, H., Kim, W. H., Kim, Y. Y., & Park, H. Y. (2020). Air pollution and central nervous system disease: A review of the impact of fine particulate matter on neurological disorders. *Frontiers in Public Health*, 8, 575330.
- Koutsyiannis, D. (2021). Rethinking climate, climate change, and their relationship with water. *Water*, 13, 849.
- Kumar, A., Bhattacharya, T., Mukherjee, S., & Sarkar, B. (2022). A perspective on biochar for repairing damages in the soil–plant system caused by climate change-driven extreme weather events. *Biochar*, 4, 22.
- Lalah JO, Otieno PO, Odira Z, Ogunah JA. Pesticides: chemistry, manufacturing, regulation, usage and impacts on population in Kenya. *Pestic. - Updat. Tox. Risk Assess. IntechOpen*. 2022.
- Lee F.-Y., Chen W.-K., Lin C.-L., Kao C.-H. Carbon monoxide poisoning and subsequent cardiovascular disease risk. *Medicine (Baltimore)* 2015;94.
- Lin, L., Yang, H., & Xu, X. (2022). Effects of water pollution on human health and disease heterogeneity: A review. *Frontiers in Environmental Science*, 10, 880246.
- Liu, H., Wang, N., Chen, D., Tan, Q., Song, D., & Huang, F. (2022). How photochemically consumed volatile organic compounds affect ozone formation: A case study in Chengdu. *China. Atmosphere (Basel)*, 13, 1534.
- Liu, M., Huan, H., Li, H., et al. (2024). Quantitative Assessment and Validation of Groundwater Pollution Risk in Southwest Karst Area. *Expo Health*. <https://doi.org/10.1007/s12403-024-00647-3>
- Maciejczyk P., Chen L.-C., Thurston G. The role of fossil fuel combustion metals in PM2.5 air pollution health associations. *Atmosphere (Basel)* 2021; 12:1086.
- Mahajan-Tatpate, P., Dhume, S., Jayakumar, N., Chendake, Y., Chavan, S., Khomane, R., & Wadagaonkar, V. (2024). Separation of Cr and Mn by polysulfone (PSF) based ultrafiltration (UF) membranes anchored with nanoparticles - Mathematical analysis of rejection properties. *Environmental Engineering Research*, 29(4), 230543.
- Manikandan, S., & Saraswathi, R. (2023). Electrocoagulation technique for removing Organic and Inorganic pollutants (COD) from the various industrial effluents: An overview. *Environmental Engineering Research*, 28(4), 220–231.
- Marques, P., Piqueras, L., & Sanz, M.-J. (2021). An updated overview of e-cigarette impact on human health. *Respiratory Research*, 22, 151.
- Mason, L. H., Harp, J. P., & Han, D. Y. (2014). Pb neurotoxicity: Neuropsychological effects of lead toxicity. *BioMed Research International*, 2014, 1–8.
- Mihalache, O.A., Elliott, C. & Dall'Asta, C. Human Health Impact Based on Adult European Consumers' Dietary Exposure to Chemical Contaminants and Consumption of Unprocessed Red Meat, Processed Meat, and Legumes. *Expo Health* (2024). <https://doi.org/10.1007/s12403-024-00634-8>
- Mishra, R. K., Mentha, S. S., Misra, Y., & Dwivedi, N. (2023). Emerging pollutants of severe environmental concern in water and wastewater: A comprehensive review on current developments and future research. *Water-Energy Nexus*, 6, 74–95.
- Montagnino E, Lytle DA, Rose J, Cwiertny D, Whelton AJ. School and childcare center drinking water: copper chemistry, health effects, occurrence, and remediation. *AWWA Water Sci*. 2022;4.
- Montes, C. M., Demler, H. J., Li, S., Martin, D. G., & Ainsworth, E. A. (2022). Approaches to investigate crop responses to ozone pollution: From O3-FACE to satellite-enabled modeling. *The Plant Journal*, 109, 432–446.
- Morantes, G., González, J. C., & Rincón, G. (2021). Characterisation of particulate matter and identification of emission sources in Greater Caracas. *Air Quality, Atmosphere & Health*, 14, 1989–2014.
- Motlagh, A. M., & Yang, Z. (2019). Detection and occurrence of indicator organisms and pathogens. *Water Environment Research*, 91, 1402–1408.
- Mughal BB, Fini J-B, Demeneix BA. Thyroid-disrupting chemicals and brain development: an update. *Endocrine Connection*. 2018;7
- Mustafa, S., Long, Y., & Rana, S. (2023). The role of corporate social responsibility and government incentives in installing industrial wastewater treatment plants: SEM-ANN deep learning approach. *Science and Reports*, 13, 16529.
- Naidu R., Biswas B., Willett I.R., Cribb J., Kumar Singh B., Nathanail C.P., Coulon F., Semple K.T., Jones K.C., Barclay A., Aitken R.J. Chemical pollution: A growing peril and potential catastrophic risk to humanity. *Environment International*. 2021; 156:106616.
- Narwal, N., Katyal, D., Kataria, N., Rose, P. K., Warkar, S. G., Pugazhendhi, A., Ghotekar, S., & Khoo, K. S. (2023). Emerging micropollutants in aquatic ecosystems and nanotechnology-based removal alternatives: A review. *Chemosphere*, 341, 139945.
- Ng, N. L., Brown, S. S., Archibald, A. T., Atlas, E., Cohen, R. C., Crowley, J. N., Day, D. A., Donahue, N. M., Fry, J. L., Fuchs, H., Griffin, R. J., Guzman, M. I., Herrmann, H., Hodzic, A., Iinuma, Y., Jimenez, J. L., Kiendler-Scharr, A., Lee, B. H., Lueken, D. J., ... Zaveri, R. A. (2017). Nitrate radicals and biogenic volatile organic compounds: Oxidation, mechanisms, and organic aerosol. *Atmospheric Chemistry and Physics*, 17, 2103–2162.
- Nguyen, V. T., Nguyen, T. B., Vo, T. D. H., Dat, N. D., Vo, T. K. Q., Nguyen, X. C., Dinh, V. C., Le, T. N. C., Duong, T. G. H., Bui, M. H., & Bui, X. T. (2024). Preliminary study of doxycycline adsorption from aqueous solution on alkaline modified biochar derived from banana peel. *Environmental Engineering Research*, 29(3), 230196.
- Nur-E-Alam, M., Mia, M. A. S., Ahmad, F., & Rahman, M. M. (2020). An overview of chromium removal techniques from tannery effluent. *Applied Water Science*, 10, 205.
- Omarova, A., Tussupova, K., Berndtsson, R., Kalishev, M., & Sharapatova, K. (2018). Protozoan parasites in drinking water: A system approach for improved water, sanitation and hygiene in developing countries. *International Journal of Environmental Research and Public Health*, 15, 495.
- Owheruo JO, Edo GI, Ifesan BO, Bolade MK, Origbemisiye BA, Akpogheli PO, Agbo JJ. Evaluation of nutraceutical

- property of extruded breakfast cereal produced from blends of malted finger millet (*Eleusine coracana*) and watermelon (*Citrullus lanatus*) seed flour. *Vegetos*. 2023.
- Pałczyński, C., Kupryś-Lipinska, I., Wittczak, T., Jassem, E., Breborowicz, A., & Kuna, P. (2018). The position paper of the Polish Society of Allergology on climate changes, natural disasters and allergy and asthma. *Advance Dermatology Allergol*, 35, 552–562.
- Parvez, F., Chen, Y., Yunus, M., Olopade, C., Segers, S., Slavkovich, V., Argos, M., Hasan, R., Ahmed, A., Islam, T., Akter, M. M., Graziano, J. H., & Ahsan, H. (2013). Arsenic exposure, impaired lung function, findings from a large population-based prospective cohort study. *American Journal of Respiratory and Critical Care Medicine*, 188, 813–819.
- Pasaoglulari Aydinlik N, Abubakar J, Edo G. Phytochemical and GCMS analysis on the ethanol extract of *Foeniculum vulgare* and *Petroselinum crispum* leaves International Journal of Chemical Technology. 2021.
- Peng, L., & Jia, R. (2023). Exploring the interplay of the physical environment and organizational climate in innovation. *Sustainability*, 15, 15013.
- Pingali, P. L. (2012). Green revolution: Impacts, limits, and the path ahead. *Proceedings of the National Academy of Sciences*, 109, 12302–12308.
- Poikane, S., Cantonati, M., Stevens, P. R., Heino, J., Turak, E., Bolpagni, R., Richardson, D. C., Borriani, A., Čtvrtlíková, M., Cid, N., Hájek, M., Galassi, D. M. P., Levkov, Z., Hawes, I., Saber, A. A., Naselli-Flores, L., Fiasca, B., Cicco, M. D., Kubečka, J., ... Segadelli, S. (2020). Characteristics, main impacts, and stewardship of natural and artificial freshwater environments: Consequences for biodiversity conservation. *Water*, 12, 260.
- Renner R. Reaction to the solution: lead exposure following partial service line replacement. *Environ. Health Perspect.* 2010;118.
- Reno A.L., Brooks E.G., Ameredes B.T. Mechanisms of heightened airway sensitivity and responses to inhaled SO₂ in asthmatics. *Environ. Health Insights* 2015; 9s1.S15671.
- Richiardi, L., Pignata, C., Fea, E., Bonetta, S., & Carraro, E. (2023). Are indicator microorganisms predictive of pathogens in water? *Water*, 15, 2964.
- Ritger, A. L., Curtis, A. N., & Chen, C. Y. (2018). Bioaccumulation of mercury and other metal contaminants in invasive lionfish (*Pterois volitans*/miles) from Curaçao. *Marine Pollution Bulletin*, 131, 38–44.
- Robb F.T., Techtman S.M. Life on the fringe: microbial adaptation to growth on carbon monoxide. *F1000Research* 2018; 7:1981.
- Roberts, W. (2021). Air pollution and skin disorders. *Int. J. Women's Dermatol.*, 7, 91–97.
- Roy, A., Mandal, M., Das, S., Kumar, M., Popek, R., Awasthi, A., Giri, B. S., Mondal, K. C., & Sarkar, A. (2024). Non-exhaust particulate pollution in Asian countries: A comprehensive review of sources, composition, and health effects. *Environmental Engineering Research*, 29(3), 230384.
- Samuel PO, Edo GI, Oloni GO, Ugbune U, Ezekiel GO, Essaghah AEA, Agbo JJ. Effects of chemical contaminants on the ecology and evolution of organisms: a review. *Chem. Ecol.* 2023;1–37.
- Sandifer, P. A., Sutton-Grier, A. E., & Ward, B. P. (2015). Exploring connections among nature, biodiversity, ecosystem services, and human health and well-being: Opportunities to enhance health and biodiversity conservation. *Ecosystem Services*, 12, 1–15.
- Saravanan, S., Davis, P. R., & Skariah, T. B. (2024). Sustainable engineered geopolymer composites: A study on the potential of fly ash, BOF slag, and iron ore tailings. *Environmental Engineering Research*, 29(5), 230540.
- Sathya, K., Nagarajan, K., GeorMalar, G. C., Rajalakshmi, S., & Lakshmi, P. R. (2022). A comprehensive review on comparison among effluent treatment methods and modern methods of treatment of industrial wastewater effluent from different sources. *Applied Water Science*, 12, 70.
- Saxena, V., Kumar, N., & Nangia, U. (2021a). Smart Grid: A Sustainable Smart Approach. *Journal of Physics: Conference Series*, 2007, 012042.
- Saxena V, Kumar N, Nangia U, Singh J. Analysis of Smart Electricity Grid Framework Unified with Renewably Distributed Generation. In: Advances in Smart Communication and Imaging Systems. Lecture Notes in Electrical Engineering, vol 721. Springer, Singapore; 2021.
- Saxena, V., Kumar, N., & Nangia, U. (2021c). Protagonist of renewable energy in distributed generation: A review. *Revista Tecnología En Marcha*, 34(4), 3–15.
- Saxena, V., Kumar, N., & Nangia, U. (2022a). Recent Trends in the Optimization of Renewable Distributed Generation: A Review. *Ingeniería e Investigación*, 42(3).
- Saxena, V., Kumar, N., & Nangia, U. (2022b). An impact assessment of distributed generation in distribution network. In *Artificial Intelligence and Sustainable Computing. Algorithms for Intelligent Systems*. Springer.
- Saxena, V., Kumar, N., & Nangia, U. (2023). An Extensive Data-Based Assessment of Optimization Techniques for Distributed Generation Allocation: Conventional to Modern. *Arch Computat Methods Eng.*, 30, 675–701.
- Saxena, V., Kumar, N., & Nangia, U. (2024a). Impact Analysis of Demand Response on the Optimal Placement of Solar PV Systems in the Distribution Network. *Brazilian Archives of Biology and Technology*, 67, e24230419.
- Saxena, V., Kumar, N., & Nangia, U. (2024b). Coal power plant-enabled grid resilience through distributed energy resources and demand response integration. *Electrical Engineering*.
- Saxena, V., Manna, S., Rajput, S. K., et al. (2024c). Sustainable energy solutions: Integrating hybrid CCHP systems with renewable technologies for efficient urban development. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 46, 668.
- Saxena, V., Manna, S., Rajput, S. K., Kumar, P., Sharma, B., Alsharif, M. H., & Kim, M. K. (2024d). Navigating the complexities of distributed generation: Integration, challenges, and solutions. *Energy Reports*, 12, 3302–3322.
- Schraufnagel, D. E. (2020). The health effects of ultrafine particles. *Experimental & Molecular Medicine*, 52, 311–317.
- Schwartz, H., Marushka, L., Chan, H. M., Batal, M., Sadik, T., Ing, A., Fediuk, K., & Tikhonov, C. (2021). Pharmaceuticals in source waters of 95 First Nations in Canada. *Canadian Journal of Public Health*, 112, 133–153.

- Schwarzenbach, R. P., Escher, B. I., Fenner, K., Hofstetter, T. B., Johnson, C. A., von Gunten, U., & Wehrli, B. (2006). The challenge of micropollutants in aquatic systems. *Science*, 313, 1072–1077.
- Sellami, A., & Elleuch, R. (2024). Green composite friction materials: A review of a new generation of eco-friendly brake materials for sustainability. *Environmental Engineering Research*, 29(3), 230226.
- Semenza JC, Ebi KL. Climate change impact on migration, travel, travel destinations and the tourism industry. *Journal of Travel Medicine* 2019;26.
- Seo, J., Rho, P., & Bae, H. K. (2023). Estimations for water quality factors of Kumho River, South Korea, using remote sensing technology. *Environmental Engineering Research*, 28(5), 220368.
- Shaji, E., Santosh, M., Sarath, K. V., Prakash, P., Deepchand, V., & Divya, B. V. (2021). Arsenic contamination of groundwater: A global synopsis with focus on the Indian peninsula. *Geoscience Frontiers*, 12, 101079.
- Sharma, S., Ahtesham, B., Muhammad, J., Lee, H., Juyeol, L., Choonsoo, P., & Duckshin, P. (2024). Improvement of indoor air quality using a smart gate that can lessen viral aerosol (MS2 virus) and particulate matter (PM): Experimental findings. *Environmental Engineering Research*, 29(2), 230269.
- Sheoran, K., Siwal, S. S., Kapoor, D., Singh, N., Saini, A. K., Alsanie, W. F., & Thakur, V. K. (2022). Air pollutants removal using biofiltration technique: A challenge at the frontiers of sustainable environment. *ACS Eng. Au.*, 2, 378–396.
- Shi, C., Yan, B., Zuo, X., Wang, C., Li, Z., & Zhu, L. (2023). The effect of aeration mode on the operational effectiveness and membrane bioreactors for greywater treatment and membrane fouling. *Environmental Engineering Research*, 28(2), 210637.
- Shi, Y., Wang, Y., Zhang, X., et al. (2024). Areal Health Risk Assessment Using Soil Bioaccessible Heavy Metal(loid)s Around Industrial Area in Nanjing. *Southeast China. Expo Health*. <https://doi.org/10.1007/s12403-024-00643>
- Siddiqua, A., Hahladakis, J. N., & Al-Attiya, W. A. K. A. (2022). An overview of the environmental pollution and health effects associated with waste landfilling and open dumping. *Environmental Science and Pollution Research*, 29, 58514–58536.
- Silva, J. A. (2023). Wastewater treatment and reuse for sustainable water resources management: A systematic literature review. *Sustainability*, 15, 10940.
- Singh MK, Saxena V. Voltage Conditioning and Harmonic Mitigation Using UPQC: A Review. In: *Innovations in Cyber Physical Systems. Lecture Notes in Electrical Engineering*, vol 788. Springer, Singapore; 2021.
- Skendžić, S., Zovko, M., Živković, I. P., Lešić, V., & Lemić, D. (2021). The impact of climate change on agricultural insect pests. *Insects*, 12, 440.
- Solomon, E. T., Gari, S. R., Kloos, H., & Alemu, B. M. (2021). Handwashing effect on diarrheal incidence in children under 5 years old in rural eastern Ethiopia: A cluster randomized controlled trial. *Tropical Medicine & International Health*, 49, 26.
- Sorensen, C., Saunik, S., Sehgal, M., Tewary, A., Govindan, M., Lemery, J., & Balbus, J. (2018). Climate change and women's health: Impacts and opportunities in India. *Geo-Health*, 2, 283–297.
- Southerland VA, Brauer M, Mohegh A, Hammer MS, van Donkelaar A, Martin RV, Apte JS, Anenberg SC. Global urban temporal trends in fine particulate matter (PM_{2.5}) and attributable health burdens: estimates from global datasets. *Lancet Planet. Heal.* 2022;6–e146.
- Srivastava, S., & Roy, T. (2023). Integrated flood risk assessment of properties and associated population at county scale for Nebraska. *USA. Sci. Rep.*, 13, 19702.
- Srivastava, S., Kumar, N., Malakar, A., Das Choudhury, S., Ray, C., & Roy, T. (2024). A machine learning-based probabilistic approach for irrigation scheduling. *Water Resour. Manag.*, 38(5), 1639–1653.
- Srivastava S, Basche A, Traylor E, Roy T. The efficacy of conservation practices in reducing floods and improving water quality. *Front. Environ. Sci.* 2023; 11.
- Steinmaus, C. M. (2016). Perchlorate in water supplies: Sources, exposures, and health effects. *Current Environmental Health Reports*, 3, 136–143.
- Syafrudin, M., Kristanti, R. A., Yuniarto, A., Hadibarata, T., Rhee, J., Al-onazi, W. A., Algarni, T. S., Almarri, A. H., & Al-Mohaimed, A. M. (2021). Pesticides in drinking water—a review. *International Journal of Environmental Research and Public Health*, 18, 468.
- Syed, M. M. M., Hossain, M. S., Karim, M. R., Uddin, M. F., Hasan, M., & Khan, R. H. (2023). Surface water quality profiling using the water quality index, pollution index and statistical methods: A critical review. *Environ. Sustain. Indic.*, 18, 100247.
- Tudi, M., Ruan, H. D., Wang, L., Lyu, J., Sadler, R., Connell, D., Chu, C., & Phung, D. T. (2021). Agriculture development, pesticide application and its impact on the environment. *International Journal of Environmental Research and Public Health*, 18, 1112.
- Tumolo, M., Ancona, V., De Paola, D., Losacco, D., Campanale, C., Massarelli, C., & Uricchio, V. F. (2020). Chromium pollution in European water, sources, health risk, and remediation strategies: An overview. *International Journal of Environmental Research and Public Health*, 17, 5438.
- Ugbune, U., Avwenaghegha, J. O., & Edo, G. I. (2023). Geochemical distribution and ecological risk level of inorganic pollutant of River Ethiopia sediment. *Dutse Journal of Pure and Applied Sciences*, 9, 187–193.
- Upadhyay, R. K. (2020). Markers for global climate change and its impact on social, biological and ecological systems: A review. *American Journal of Climate Change*, 9, 159–203.
- Vardoulakis, S., Dimitroulopoulou, C., Thornes, J., Lai, K.-M., Taylor, J., Myers, I., Heaviside, C., Mavrogianni, A., Shrubsole, C., Chalabi, Z., Davies, M., & Wilkinson, P. (2015). Impact of climate change on the domestic indoor environment and associated health risks in the UK. *Environment International*, 85, 299–313.
- Waite T, Murray V, Baker D. Carbon monoxide poisoning and flooding: changes in risk before, during and after flooding require appropriate public health interventions. *PLoS Curr.* 2014..
- Wallington, T. J., Anderson, J. E., Dolan, R. H., & Winkler, S. L. (2022). Vehicle emissions and urban air quality: 60 years of progress. *Atmosphere (Basel)*, 13, 650.

- Wang, Y., & Qian, H. (2021). Phthalates and their impacts on human health. *Healthcare*, 9, 603.
- Wang, M., Janssen, A. B. G., Bazin, J., Stokal, M., Ma, L., & Kroeze, C. (2022). Accounting for interactions between sustainable development goals is essential for water pollution control in China. *Nature Communications*, 13, 730.
- Wang, H., Dai, J., Chen, H., Wang, F., Zhu, Y., Liu, J., Zhou, B., & Yuan, R. (2023). Adsorption of phosphate by Mg/Fe-doped wheat straw biochars optimized using response surface methodology: Mechanisms and application in domestic sewage. *Environmental Engineering Research*, 28(2), 210602.
- Wani, A. L., Ara, A., & Usmani, J. A. (2015). Lead toxicity: A review. *Interdisciplinary Toxicology*, 8, 55–64.
- Ward, M., Jones, R., Brender, J., de Kok, T., Weyer, P., Nolan, B., Villanueva, C., & van Breda, S. (2018). Drinking water nitrate and human health: An updated review. *International Journal of Environmental Research and Public Health*, 15, 1557.
- Wasielak-Politowska, M., & Kordowitzki, P. (2022). Chromosome segregation in the oocyte: What goes wrong during aging. *International Journal of Molecular Sciences*, 23, 2880.
- Wei, X., Cai, S., Ni, P., & Zhan, W. (2020). Impacts of climate change and human activities on the water discharge and sediment load of the Pearl River, southern China. *Science and Reports*, 10, 16743.
- Weiland-Bräuer, N. (2021). Friends or foes—microbial interactions in nature. *Biology (Basel)*, 10, 496.
- White, B. P., Breakey, S., Brown, M. J., Smith, J. R., Tarbet, A., Nicholas, P. K., & Ros, A. M. V. (2023). Mental health impacts of climate change among vulnerable populations globally: An integrative review. *Annals of Global Health*, 89, 66.
- Wikel, S. K. (2022). Changing geographic ranges of human biting ticks and implications for tick-borne zoonoses in North America. *Zoonotic Diseases*, 2, 126–146.
- Wróblewska, K., & Jeong, B. R. (2021). Effectiveness of plants and green infrastructure utilization in ambient particulate matter removal. *Environmental Sciences Europe*, 33, 110.
- Xiaoju, L., Chuah, A. L., Shafreeza, S., Md, S. M. S., Husain, S. A., Aun, T. P., & Jinzhao, H. (2024). Variation characteristics and coordinated emission reduction of air pollutants in megacity of Chengdu-Chongqing economic circle under dual carbon goal. *Environmental Engineering Research*, 29(4), 230475.
- Xie, Y., Liu, X., Wei, H., Chen, X., Gong, N., Ahmad, S., Lee, T., Ismail, S., & Ni, S. Q. (2022). Insight into impact of sewage discharge on microbial dynamics and pathogenicity in river ecosystem. *Science and Reports*, 12, 6894.
- Yang, Z., Long, Y., Yang, X., Liu, J., & Zhu, G. (2024a). Preparation and application of polymeric silicate coagulant: A short review. *Environmental Engineering Research*, 29(5), 230672.
- Yang, K., Hong, X., Wang, X., Zhu, Y., Zuo, P., & Gao, G. (2024b). Characteristics and risk assessment of atmospheric PM_{2.5} heavy metals pollution near coal gangue sites in Huaibei, China. *Environmental Engineering Research*, 29(5), 230720–0.
- Zazouli MA, Asghari S, Tarrahi R, Lisar SYS, Babanezhad E, Dashtban N. The potential of common duckweed (*Lemna minor*) in phytoremediation of phenanthrene and pyrene. *Environmental Engineering Research*. 2023; 28(1):
- Zendehbad, M., Mostaghelchi, M., Mojganfar, M., Cepuder, P., & Loiskandl, W. (2022). Nitrate in groundwater and agricultural products: Intake and risk assessment in northeastern Iran. *Environmental Science and Pollution Research*, 29, 78603–78619.
- Zhang, Y., Yang, F., Wang, Y., Zheng, Y., & Zhu, J. (2023). Effects of acid rain stress on the physiological and biochemical characteristics of three plant species. *Forests*, 14, 1067.
- Zhang, Y., Li, L., Song, B., et al. (2024). Comprehensive Risk Assessment of Heavy Metals Based on Pollution Sources: A Case Study for Agricultural Soils from Southwest China. *Expo Health*. <https://doi.org/10.1007/s12403-024-00642>

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